# Supporting Information: Computational Discovery of Antimalarial Candidates from African Natural Product Chemical Space

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# Introduction

This supplementary document details methodological protocols, scoring metrics, and dataset statistics supporting the main text (Temgoua *et al.*, 2026). The main manuscript describes a computational framework for antimalarial drug discovery that integrates African natural products with synthetic antimalarials, identifies antimalarial candidates (including select multi-target hits) through dual-filter consensus docking, and prioritizes leads using multi-parameter optimization scoring.

The supplementary content elaborates the Multi-Parameter Optimization (MPO) scoring framework introduced in the main manuscript (Section 3.6), presenting mathematical formulations for the five desirability components and the polypharmacology bonus. Tabular summaries detail chemical space analysis, scaffold diversity, and drug-likeness characteristics of the 65 856-molecule hybrid library. Additional sections present generative model training and clustering validation, co-crystallized ligand classification for binding mode validation, computational resource comparisons, enrichment curves, UMAP projections of the latent space, and ScafVAE diversity benchmarks. Final tables provide target-specific docking outcomes, selectivity indices, and cytochrome P450 inhibition profiles for the 810 screened seed molecules with valid selectivity predictions against three *Plasmodium falciparum* structures: dihydrofolate reductase (DHFR, PDB: 7F3Y), chloroquine resistance transporter (CRT, 6UKJ), and ATPase 4 (ATP4, 9N10).

Calculation results referenced in this supplementary material are available from the project repository (`https://github.com/NanaEngo/Malaria_codesV2`) together with the analysis scripts in the `scripts/` directory.

## Multi-parameter optimization scoring function

The MPO framework integrates five normalized desirability components into a composite score for hit prioritization. This approach extends the single-objective docking optimization strategy by simultaneously considering binding affinity, geometric confidence, drug-likeness, ADMET safety, and structural rule compliance. The methodology builds upon established multi-parameter optimization principles adapted for computational drug discovery applications.[S1,S2]

The total MPO score for compound $i$ against target $j$ is defined as:

$$\text{MPO}_{\text{total},i} = \sum_{k=1}^{5} w_k \cdot S_{k,i} + \text{Polypharmacology Bonus}_i \tag{1}$$

where $w_k$ are the component weights (Vina 35 %, DiffDock 25 %, QED 20 %, ADMET 15 %, Ro5 5 %) and $S_{k,i}$ are the normalized desirability scores scaled to [0, 1]. The polypharmacology bonus rewards compounds meeting the computational score threshold against multiple targets simultaneously; it does not establish biological polypharmacology.

### Weight selection rationale

The component weights reflect established priorities in early-stage computational drug discovery.[S1,S2] The 35 % weight assigned to the Vina affinity score prioritizes the physics-inspired docking signal as the primary filter; this score is not a measured thermodynamic quantity. DiffDock geometric confidence (25 %) provides a complementary computational screening signal,[S3] with the combined 60 % weight for docking components reflecting the central role of target engagement in hit identification.

Drug-likeness (QED, 20 %) receives substantial weight to ensure compounds satisfy oral bioavailability criteria,[S4] while ADMET safety (15 %) addresses early-stage toxicity concerns.[S5] The modest 5 % weight for Lipinski violations reflects that Ro5 compliance is already captured partially within QED, and modern drug discovery increasingly tolerates single violations for complex targets.[S1] This weighting scheme balances potency, developability, and safety considerations typical of lead identification campaigns.[S2]

## Component 1: Vina binding affinity ($S_{\mathbf{vina}}$, 35 %)

The Vina component is linearly scaled between the observed minimum and maximum binding affinities across all docked compounds:

$$S_{\text{vina},i} = \frac{\text{Vina}_i - \text{Vina}_{\text{min}}}{\text{Vina}_{\text{max}} - \text{Vina}_{\text{min}}} \tag{2}$$

where $\text{Vina}_i$ is the best binding affinity (kcal mol$^{-1}$) for compound $i$, and $\text{Vina}_{\text{min}}$, $\text{Vina}_{\text{max}}$ are the dataset bounds. More negative values yield higher desirability. This linear transformation preserves the rank ordering of compounds while mapping raw energies to a [0, 1] scale compatible with other desirability components. Linear scaling was selected over exponential or power-law transformations to maintain proportional differences in binding affinity across the scoring range, consistent with the approximately linear relationship between binding free energy and experimental potency within the typical docking score range of $-10$ kcal mol$^{-1}$ to $-5$ kcal mol$^{-1}$.

## Component 2: DiffDock confidence ($S_{\mathbf{diff}}$, 25 %)

The DiffDock confidence score is transformed via a logistic sigmoid function to map structural log-odds to probabilistic desirability:

$$S_{\text{diff},i} = \frac{1}{1 + e^{-c_i}} \tag{3}$$

where $c_i$ is the DiffDock confidence score for compound $i$.[S3] This transformation ensures that high-confidence poses ($c > 0$) receive scores approaching 1.0, while low-confidence poses ($c < -1.5$) receive scores approaching 0. The sigmoidal mapping was chosen over linear scaling because DiffDock confidence scores follow a non-uniform distribution with a sharp transition around $c = 0$, and the sigmoid function naturally captures this threshold behavior while providing smooth gradation for intermediate confidence values.

## Component 3: Quantitative estimate of drug-likeness ($S_{\mathbf{qed}}$, 20 %)

The QED score is used directly as a desirability component without transformation:

$$S_{\mathrm{qed},i} = \mathrm{QED}_i \tag{4}$$

where $\mathrm{QED}_i \in [0, 1]$ is the quantitative estimate of drug-likeness for compound $i$, computed from eight molecular properties using the RDKit implementation.[S6] QED values $> 0.70$ are considered indicative of drug-like compounds.[S4]

## Component 4: ADMET safety profile ($S_{\mathbf{admet}}$, 15 %)

The ADMET component aggregates 11 critical pharmacokinetic and safety parameters into a composite desirability index:

$$S_{\mathrm{admet},i} = \frac{1}{11} \sum_{m=1}^{11} d_m(\mathrm{ADMET}_{i,m}) \tag{5}$$

where $d_m$ is the individual desirability function for ADMET endpoint $m$, predicted using ADMET-AI.[S5] The 11 endpoints are:

- Oral bioavailability (human intestinal absorption, HIA)
- Aqueous solubility ($\log S$)
- Caco-2 permeability
- Blood-brain barrier penetration
- hERG liability (cardiotoxicity risk)

- AMES mutagenicity
- Hepatotoxicity (DILI)
- Synthetic accessibility (SA score)
- P-glycoprotein substrate status
- Cytochrome P450 inhibition risk (composite of CYP2C9, 2C19, 3A4, 2D6)
- Total clearance profile

Each endpoint is scored as a binary or continuous desirability function (1 = favourable, 0 = unfavourable), with thresholds derived from FDA-approved drug distributions.

## Component 5: Lipinski's rule of five penalty ($P_{\mathbf{Ro5}}$, 5 %)

A penalty term proportional to formal Lipinski's rule of five violations:[S4]

$$P_{\text{Ro5},i} = 1 - 0.2 \cdot n_{\text{violations},i} \tag{6}$$

where $n_{\text{violations},i}$ is the number of Ro5 violations (0 to 4). The Ro5 criteria are: molecular weight $> 500\,\text{Da}$, $\log P > 5$, hydrogen bond donors $> 5$, and hydrogen bond acceptors $> 10$. Penalties are: 0 violations = 1.0, 1 violation = 0.8, 2 violations = 0.6, 3 violations = 0.4, 4 violations = 0.2.

## Polypharmacology bonus

Compounds exhibiting binding ($\text{MPO}_{\text{target}} \geq 0.50$) against multiple targets receive an additional bonus to reward multi-target activity:

$$\text{Polypharmacology Bonus}_i = 0.05 \cdot (n_{\text{targets},i} - 1) \quad \text{if } n_{\text{targets},i} \geq 2 \tag{7}$$

where $n_{\text{targets},i}$ is the number of targets against which compound $i$ achieves $\text{MPO}_{\text{target}} \geq 0.50$. This bonus is capped at 0.10 (3 targets). The 0.05 increment per additional target represents a modest computational preference for multi-target profiles that may be relevant to resistance hypotheses; it does not demonstrate resistance circumvention or therapeutic advantage. The MPO $\geq 0.50$ threshold ensures that only compounds meeting minimum quality criteria across all scoring components qualify for the bonus, thereby distinguishing genuine multi-target candidates from non-selective binders with poor drug-likeness or safety profiles.

## Centroid-based screening and MPO stratification

The MPO framework was applied within a centroid-based virtual screening workflow to enable efficient exploration of the 65 856-molecule library. Rather than exhaustively docking all compounds, the workflow proceeded as follows:

1. **K-Means clustering:** The library was partitioned into 484 clusters based on 64-dimensional VAE latent representations, with one centroid selected per cluster to represent its chemical neighborhood.

2. **Consensus docking:** The 484 centroids were docked against three *P. falciparum* targets (PfDHFR, PfCRT, and PfATP4) using dual-filter consensus scoring (AutoDock Vina + DiffDock). Centroids were classified as EXCELLENT, GOOD, or OTHERS based on binding affinity and geometric confidence thresholds. Failed records were excluded from consensus scoring and MPO calculation; target-specific denominators are reported with the corresponding results.

3. **MPO scoring:** 93 centroids passing consensus classification (EXCELLENT or GOOD) received full MPO evaluation across all five desirability components (Vina, DiffDock, QED, ADMET, Ro5).

4. **Optimization-worthy threshold:** 76 centroids achieved MPO $\geq 0.40$, defining the

optimization-worthy tier. Within this tier, 7 centroids achieved MPO $\geq$ 0.50 (secondary hits), while 69 centroids fell in the 0.40 to 0.50 range (promising scaffolds).

5. **Scaffold expansion:** The 76 optimization-worthy centroids map to 53 unique clusters (multiple centroids can represent the same cluster because centroids are selected as representative molecules rather than cluster identifiers). Scaffold expansion from these 53 clusters yielded 40 481 molecules (mean cluster size: 763.8) for downstream lead optimization. Of these, 19 913 molecules satisfied the final synthetic accessibility and MPO criteria, defining the final set of synthesizable leads.

This centroid-based strategy reduced computational cost by 99.3 % compared to exhaustive library-scale docking (Table S33) while preserving chemical diversity through cluster-based sampling. The MPO $\geq$ 0.40 threshold was selected to balance stringency with practical hit list size, ensuring that expanded clusters represent chemically tractable starting points for medicinal chemistry optimization.

## Natural product relatedness filtering

To identify candidates retaining structural features of African natural products, NP-relatedness was assessed using Tanimoto similarity to 396 seed African NPs. Similarity was computed using Morgan fingerprints (radius 2, 2 048 bits),[S7] with compounds achieving maximum Tanimoto $\geq$ 0.4 to any seed NP classified as NP-related. This threshold balances scaffold conservation with sufficient structural diversity for lead optimization.

Among the 76 optimization-worthy centroids, 13 (17.1 %) satisfied the NP-relatedness criterion (Tanimoto $\geq$ 0.4), representing candidates that combine favorable multi-target profiles with natural product-inspired scaffolds. The remaining 63 centroids (82.9 %) represent structurally distinct analogues with Tanimoto $<$ 0.4, confirming that generative expansion explored new chemical space while preserving drug-like characteristics.

This dual population supports two computational lead-optimization strategies: NP-related candidates ($n = 13$) retain similarity to annotated seed scaffolds, whereas structurally distinct candidates ($n = 63$) expand the explored chemical space. Neither group establishes biological activity, toxicity, or resistance circumvention. The NP-relatedness filter was applied post-MPO scoring.

## Framework limitations and sensitivity

**Sensitivity methodology.** The five MPO component scores comprise $S_{vina}$ (Vina binding affinity), $S_{diff}$ (DiffDock confidence), $S_{qed}$ (QED drug-likeness), $S_{admet}$ (ADMET composite), and $P_{Ro5}$ (Lipinski penalty). Because individual Vina and DiffDock scores are not stored per molecule in the c6 library, $S_{vina}$ and $S_{diff}$ are back-calculated from the shared residual of the weighted MPO score after subtracting the QED, ADMET, and Ro5 contributions, then split proportionally (35:25 ratio). Weight perturbations of Vina and DiffDock therefore reallocate within this combined residual rather than varying the two components independently. QED, ADMET, and Ro5 perturbations reflect true independent variation.

The MPO framework represents a pragmatic balance between computational tractability and multi-objective optimization. Component weights were selected based on established drug discovery priorities[S1,S4] rather than target-specific optimization, which would require experimental validation data currently unavailable for the majority of the library. All docking CSVs report provenance fields (target, exhaustiveness, grid version) so that every score can be traced to the receptor preparation and Vina search depth used. The framework's primary limitation is the dataset-dependent nature of min-max scaling for Vina scores, which precludes direct score comparison across different compound libraries without renormalization.

Formal sensitivity analysis across 25 weight perturbations ($\pm 10\,\%$ to $20\,\%$ per component) yielded mean Spearman rank correlation $\rho = 0.792$ for the top-1,000 compounds and mean Jaccard similarity of 0.548 for the top-20 hit list, indicating moderate sensitivity to weight choice (Table S19). When both stability criteria (top-1000 $\rho > 0.95$ and top-20 Jaccard $\geq 0.70$) are required, only 5 % of non-baseline perturbations passed (DiffDock 25 %; Vina, QED, ADMET, and Ro5 0 %), confirming that the ranking is moderately rather than strongly robust to weight choice; a substantial fraction of perturbations did not pass the stability test. The MPO $\geq 0.40$ threshold for optimization-worthy centroids was selected to balance stringency with practical cluster expansion size, yielding 76 centroids that map to 53 unique clusters and expand to 40 481 molecules (19 913 final synthesizable leads) representing chemically tractable starting points for medicinal chemistry optimization.

# Chemical space analysis and drug-likeness tables

The following tables summarize the chemical diversity, drug-likeness characteristics, and target-specific docking outcomes for the 65 856-molecule hybrid library.

Table S1: Top 10 most prevalent Bemis–Murcko scaffolds present in the 65 856-molecule generative hybrid library, highlighting the structural emphasis on privileged aromatic systems. *The library is dominated by benzene and quinoline-like privileged scaffolds; structural compatibility with antimalarial pharmacophores requires experimental or target-specific validation.*

| Rank | Scaffold SMILES | Count | % | Description |
|---|---|---|---|---|
| 1 | c1ccccc1 | 5 537 | 8.4 % | Benzene ring |
| 2 | (acyclic) | 3 610 | 5.5 % | Acyclic compounds |
| 3 | c1ccc2ncncc2c1 | 1 644 | 2.5 % | Quinazoline core |
| 4 | c1ccc(CCNCc2ccccc2)cc1 | 1 441 | 2.2 % | Diphenylethylamine |
| 5 | O=C(C=Cc1ccccc1)c1ccccc1 | 1 315 | 2.0 % | Chalcone scaffold |
| 6 | c1ccc2ncccc2c1 | 1 308 | 2.0 % | Quinoline core |
| 7 | O=c1cc[nH]c2ccccc12 | 1 176 | 1.8 % | Quinolin-2-one |
| 8 | c1ccc(Nc2ccnc3ccccc23)cc1 | 823 | 1.2 % | Phenylquinoline amine |
| 9 | O=c1ccc2ccccc2o1 | 798 | 1.2 % | Coumarin scaffold |
| 10 | O=C(Nc1ccccc1)c1ccccc1 | 787 | 1.2 % | Benzamide derivative |
| **Summary:** 20 702 unique scaffolds, 31.4 % diversity, Top 10 = 28.0 % of library | | | | |

Table S2: Distribution of predicted selectivity index proxies ($SI_{pred} = score_{eos7kpb}/(1 - DILI_{ADMET\text{-}AI})$) for 810 screened seed molecules with valid predictions. This computational proxy approximates antiparasitic selectivity over mammalian cytotoxicity; experimental $IC_{50}$(HepG2)/$IC_{50}$(Pf NF54) determination is required for confirmation. *All screened molecules with valid SI predictions exceed the $SI_{pred} > 10$ proxy threshold; however, $SI_{pred}$ is an unvalidated prioritization heuristic (sensitive to the DILI denominator, with an observed maximum of 106 773.8) and does not by itself establish a therapeutic window, which requires experimental $IC_{50}(HepG2)/IC_{50}(Pf\ NF54)$ determination.*

| Metric | Value |
|---|---|
| Total molecules screened | 810 |
| Selectively antiparasitic (SI > 10) | 100 % (810) |
| Mean selectivity index | 993.8 |
| Median selectivity index | 283.5 |
| Maximum selectivity index | 106 773.8 |

Table S3: Cytochrome P450 (CYP) inhibition interaction risks for the 810 screened seed molecules with valid predictions, identifying moderate metabolic interaction probabilities (threshold $> 0.5$) relevant for formulation and drug-drug interactions. *Metabolic risk profiles are isoform-dependent, with CYP2C19 and CYP3A4 exhibiting the highest interaction probabilities.*

| **CYP Isoform** | **Mean** | **Median** | **High Inhibitors** | **%** |
|---|---|---|---|---|
| CYP2C9 | 0.261 | 0.082 | 207 | 24.4 % |
| CYP2C19 | 0.424 | 0.346 | 356 | 41.9 % |
| CYP3A4 | 0.426 | 0.338 | 361 | 42.5 % |
| CYP2D6 | 0.170 | 0.001 | 138 | 16.2 % |

Table S4: Stage-specific antimalarial activity predictions derived from the eos80ch model, indicating preferential predicted activity against asexual blood-stage parasites compared with transmission-blocking sexual gametocyte stages. *Identified candidates are predicted to primarily target the asexual blood stage, consistent with the clinical requirements for acute malaria treatment; these are computational predictions requiring experimental confirmation.*

| **Stage** | **Median** | **Mean** | **Interpretation** |
|---|---|---|---|
| Asexual blood stage | 0.567 | 0.647 | Preferred predicted stage |
| Sexual stage (gametocytes) | 0.244 | 0.502 | Transmission-blocking potential |

Table S5: Explained variance ratio across the first four principal components derived from PCA of nine standardized physicochemical descriptors (MW, logP, HBA, HBD, TPSA, QED, SA score, rotatable bonds, stereo centers). Components are ordered by decreasing explained variance (PC1 $\geq$ PC2 $\geq$ PC3 $\geq$ PC4). The first three components capture 82.3 % of total structural variance. *Physicochemical diversity is highly compressible, with three components sufficing to capture the majority of library variance.*

| **Component** | **Variance** | **% Explained** | **Cumulative %** |
|---|---|---|---|
| PC1 | 0.517 | 51.7 % | 51.7 % |
| PC2 | 0.176 | 17.6 % | 69.3 % |
| PC3 | 0.131 | 13.1 % | 82.3 % |
| PC4 | 0.068 6 | 6.86 % | 89.19 % |
| **PC1–3 total:** 82.3 % of variance explained | | | |

Table S6: Recovery rate of annotated African natural product (NP) scaffolds within the final expanded generative library, summarizing the retention of privileged bioactive cores throughout the STONED-SELFIES and Cheese API expansion steps. *The observed scaffold recovery is consistent with retention of some seed-framework features, but does not establish preservation of pharmacophore function.*

| **Metric** | **Value** |
|---|---|
| Seed NP scaffolds | 101 |
| Library scaffolds | 20 702 |
| Recovered scaffolds | 70 |
| Recovery rate | 69.3 % |

Table S7: African natural product-related optimization candidates identified through centroid-based screening. NP-relatedness defined as Tanimoto $\geq$ 0.4 to 396 African NP seeds using Morgan fingerprints (radius 2, 2048 bits). *NP-related candidates span the full medicinal chemistry optimization spectrum, from secondary hits ready for lead optimization to promising scaffolds suitable for hit-to-lead development.*

| **Category** | **Total** | **NP-Related** | **Percentage** | **Mean Tanimoto** |
|---|---|---|---|---|
| Secondary Hits (MPO $\geq$ 0.50) | 7 | 2 | 28.6 % | 0.52 |
| Promising Scaffolds (MPO 0.40–0.50) | 69 | 11 | 15.9 % | 0.48 |
| **All Optimization Candidates** | **76** | **13** | **17.1 %** | **0.49** |

Table S8: Distribution metrics for the fraction of $sp^3$ hybridized carbons ($fsp^3$), providing a quantitative measure of the three-dimensional structural complexity and saturation of the hybrid library compared to purely flat aromatic synthetic compounds. *The hybrid library maintains a balanced $fsp^3$ distribution, ensuring both structural complexity and synthetic tractability.*

| **Metric** | **Value** |
|---|---|
| Mean $fsp^3$ | 0.298 |
| Median $fsp^3$ | 0.286 |
| Standard deviation | 0.172 |

[a]Final synthesisable lead count (19,913) is taken from the canonical corrected-grid output 'results/c6__primary__leads__synthesisable.csv' and reflects SYBA > 0 plus the MPO ≥ 0.40 centroid filter; it supersedes the earlier 28,679 estimate.

| **Metric** | **Pass** | **Threshold** | **Rate** |
|---|---|---|---|
| **Full Library (65 856 molecules):** | | | |
| Lipinski Ro5 | 61 975 / 65 856 | All 4 criteria | 94.1 % |
| Veber (rotatable bonds) | 58 304 / 65 856 | ≤ 10 | 88.5 % |
| SYBA > 0 | 37 562 / 65 856 | Synthetic accessibility | 57.0 % |
| NPL > 0 | 28 014 / 65 856 | NP-likeness | 42.5 % |
| SA < 3.5 | 32 804 / 65 856 | Synthetic tractability | 49.8 % |
| QED > 0.70 | 24 144 / 65 856 | Drug-likeness | 36.7 % |
| **Centroid-Based Screening Cascade:** | | | |
| K-Means centroids docked | 484 / 484 | All clusters | 100 % |
| Consensus classification pass | 93 / 484 | EXCELLENT or GOOD | 19.2 % |
| Optimization-worthy (MPO ≥ 0.40) | 76 / 93 | Centroid MPO threshold | 81.7 % |
| Secondary hits (MPO ≥ 0.50) | 7 / 76 | High MPO centroids | 9.2 % |
| **Scaffold Expansion from 53 Unique Clusters:** | | | |
| Total molecules in expansion | 40 481 | Mean cluster size: 763.8 | — |
| Final synthesizable leads[a] | 19 913 / 40 481 | SYBA > 0 + MPO filter | 49.2 % |

Table S10: Selected docking-predicted hits, including predicted multi-target candidates and single-target leads. These labels are computational predictions and do not establish biological activity.

| **Ligand** | **Target** | **Vina (kcal mol$^{-1}$)** | **DiffDock Conf.** | **Notes** |
|---|---|---|---|---|
| 201 | PfDHFR | −8.49 | −0.30 | Primary consensus lead |
| 214–PfCRT | PfCRT | −8.33 | −0.42 | Named parent lead; MD system |
| 87 | PfDHFR | −8.38 | −0.40 | Primary consensus lead |
| 438 | PfATP4 | −5.86 | −0.95 | Single `GOOD` hit |
| 440 | Multi-target | < −6.0 | > −1.0 | Polypharmacological |
| 169 | Multi-target | < −6.0 | > −1.0 | Polypharmacological |
| 161 | Multi-target | < −6.0 | > −1.0 | Polypharmacological |
| **Predicted multi-target candidates in the expanded library:** > 70 compounds meeting the declar | | | | |

Table S11: Top-20 compounds ranked by multi-parameter optimization (MPO) score. All compounds achieve perfect Lipinski compliance (4/4 criteria), high synthetic accessibility (SA < 2.7), and were generated via the STONED-SELFIES arm of the generative expansion. SMILES strings are provided for reproducibility. *MPO prioritization identifies candidates with favorable predicted drug-likeness, docking-score, and synthetic-accessibility profiles; these computational estimates require experimental validation.*

| Rank | SMILES | MPO | QED | SA | SYBA | Source |
|---|---|---|---|---|---|---|
| 1 | `Cc1ccc(CN2CCN(Cc3ccccc3)CC2)cc1O` | 0.829 | 0.939 | 1.72 | 109.01 | STONED |
| 2 | `Cc1ccccc1CN1CCN(Cc2cccc(O)c2)CC1` | 0.828 | 0.939 | 1.76 | 124.91 | STONED |
| 3 | `Cc1cc(CN2CCN(Cc3ccccc3)CC2)ccc1O` | 0.826 | 0.939 | 1.71 | 100.63 | STONED |
| 4 | `COc1ccc(CN2CCN(Cc3ccccc3C)CC2)cc1O` | 0.826 | 0.916 | 1.84 | 133.75 | STONED |
| 5 | `Oc1cccc(CN2CCN(Cc3ccc(Cl)cc3)CC2)c1` | 0.826 | 0.937 | 1.71 | 116.03 | STONED |
| 6 | `Cc1ccc(CN2CCN(Cc3ccc(O)cc3)CC2)c(C)c1` | 0.826 | 0.938 | 1.80 | 122.03 | STONED |
| 7 | `COc1ccc(CN2CCN(Cc3ccc(C)c(O)c3)CC2)cc1` | 0.826 | 0.916 | 1.80 | 119.14 | STONED |
| 8 | `Cc1ccc(CN2CCN(Cc3cccc(O)c3)CC2)c(C)c1` | 0.825 | 0.938 | 1.85 | 129.53 | STONED |
| 9 | `Oc1ccc(CN2CCN(Cc3ccccc3)CC2)cc1Cl` | 0.825 | 0.937 | 1.72 | 113.80 | STONED |
| 10 | `Oc1ccc(CN2CCN(Cc3ccccc3Cl)CC2)cc1` | 0.824 | 0.937 | 1.71 | 122.53 | STONED |
| 11 | `COc1cccc(CN2CCN(Cc3ccc(C)c(O)c3)CC2)c1` | 0.824 | 0.916 | 1.85 | 120.90 | STONED |
| 12 | `Cc1ccc(CN2CCN(Cc3cc(O)ccc3C)CC2)cc1` | 0.823 | 0.938 | 1.82 | 106.23 | STONED |
| 13 | `Oc1cccc(CN2CCN(Cc3ccccc3Cl)CC2)c1` | 0.823 | 0.937 | 1.77 | 128.98 | STONED |
| 14 | `COc1ccc(CN2CCN(Cc3ccc(O)c(C)c3)CC2)cc1` | 0.823 | 0.916 | 1.79 | 110.76 | STONED |
| 15 | `Oc1ccc(CN2CCN(Cc3ccccc3)CC2)c(Cl)c1` | 0.823 | 0.937 | 1.76 | 113.45 | STONED |
| 16 | `COc1ccc(CN2CCN(Cc3cccc(O)c3C)CC2)cc1` | 0.822 | 0.916 | 1.87 | 116.50 | STONED |
| 17 | `Oc1ccc(CN2CCN(Cc3cccc(Cl)c3)CC2)cc1` | 0.822 | 0.937 | 1.71 | 115.24 | STONED |
| 18 | `COc1ccc([C@@H]2CC(=O)c3c(O)cc(OC)cc3O2)cc1` | 0.822 | 0.943 | 2.62 | 41.36 | STONED |
| 19 | `Oc1ccc(CN2CCN(Cc3ccc(Cl)cc3)CC2)cc1` | 0.821 | 0.937 | 1.65 | 87.89 | STONED |
| 20 | `COc1cccc(CN2CCN(Cc3ccc(O)c(C)c3)CC2)c1` | 0.821 | 0.916 | 1.84 | 112.52 | STONED |

**Summary:** Mean MPO = 0.825 ± 0.003; Mean QED = 0.931 ± 0.010; Mean SA = 1.81 ± 0.20
**Drug-likeness:** 20/20 compounds QED > 0.7; 20/20 compounds SA < 3.5; 20/20 compounds SYBA > 0

Table S12: Constituent origins and corresponding molecule counts for the parallel generative expansion arms that assemble the final deduplicated 65 856-molecule hybrid antimalarial library. *The hybrid library combines similarity-based and perturbation-based expansion domains; the resulting computational profiles require experimental assessment.*

| **Source** | **File** | **Molecules** | **Description** |
|---|---|---|---|
| REINVENT scaffolds | `eos57bx_generated_mol2molscaf.csv` | 111 614 | Novel scaffolds |
| COCONUT intermediates | `eos6ost_np_mol_scaf_smi_filt.csv` | 172 750 | NP scaffold expansion |
| Final hybrid library | `eos80ch_malaria_final_activity.csv` | 65 856 | Deduplicated, filtered library |

**Note:** The counts for REINVENT scaffolds and COCONUT intermediates describe the raw generated candidate pools and intermediate datasets at intermediate stages of the workflow. They do not represent disjoint partitions of the library, and their sum does not equal the final deduplicated, filtered hybrid library size (65 856 molecules).

Table S13: Summary of cumulative explained variance for the Principal Component Analysis (PCA) used to characterize the 64-dimensional chemical space representation. *Low-dimensional representations capture the majority of structural variance, validating the use of compressed latent spaces for chemical space mapping.*

| **Metric** | **Value** |
|---|---|
| PC1–PC3 cumulative variance | 82.3 % |
| Number of molecular descriptors | 200 |
| Number of compounds analyzed | 65 856 |

Table S14: Tanimoto structural similarity evaluation comparing generated analogues against seed African natural products. Novelty was assessed using Morgan fingerprints (radius 2, 2 048 bits), defining structural novelty rigorously as a maximum Tanimoto coefficient $< 0.4$ relative to any foundational seed NP. *Over 92 % of the library represents structurally new chemical space, showing substantial separation from the evaluated natural-product reference set.*

| **Metric** | **Whole-Molecule** | **Scaffold-Only** |
|---|---:|---:|
| Molecules analyzed | 5 000 (random sample) | 5 000 (same sample) |
| Seed NPs compared against | 396 | 396 |
| Novel molecules (Tanimoto $< 0.4$) | 92.6 % (4 630) | — |
| Mean max Tanimoto | 0.206 | 0.379 |
| Std dev max Tanimoto | 0.125 | 0.308 |
| Median max Tanimoto | 0.171 | 0.262 |

**Whole-Molecule Tanimoto Distribution:**

$< 0.2$ (highly novel): 62.5 % (3 127)
0.2 to 0.3 (novel): 25.2 % (1 258)
0.3 to 0.4 (moderately novel): 4.9 % (245)
0.4 to 0.5 (similar): 3.4 % (171)
$\geq 0.5$ (very similar): 4.0 % (199)

Table S15: Pearson correlation coefficient ($r$) analysis comparing DiffDock geometric confidence scores against AutoDock Vina affinity scores, characterizing the association between the two screening dimensions across the three evaluated targets. *The observed low-to-moderate associations support using Vina and DiffDock as complementary screening dimensions, while not proving independence or eliminating scoring bias.*

| **Target (PDB)** | $n$ **ligands** | **Pearson** $r$ | $p$**-value** |
|---|---|---|---|
| PfDHFR (7F3Y) | 280 | 0.327 | $2.11 \times 10^{-8}$ |
| PfCRT (6UKJ) | 64 | 0.129 | $3.08 \times 10^{-1}$ |
| PfATP4 (9N10) | 447 | 0.113 | $1.67 \times 10^{-2}$ |

# Benchmark studies

Table S16: Enrichment analysis: MMV Malaria Box positive-control (Vina+DiffDock consensus) and DEKOIS PfDHFR independent docking (Vina only). *For two targets, the MMV comparison labels the top and bottom score strata and therefore measures retrodictive consistency rather than external discrimination. DEKOIS uses property-matched decoys docked with AutoDock Vina alone (no ML rescoring) as the independent baseline.*

| Target (PDB) | ROC-AUC | EF1 % | EF5 % | EF10 % | BEDROC | PR-AUC | Actives | Decoys |
|---|---|---|---|---|---|---|---|---|
| PfDHFR-TS (7F3Y) | 0.924 | 2.700 | 2.570 | 2.720 | 1.309 | 0.842 | 399 | 399 |
| PfATP4 (9N10) | 1.000 | 2.048 | 2.000 | 2.000 | 1.810 | 1.000 | 99 | 99 |
| PfCRT (6UKJ) | 0.971 | 1.110 | 1.110 | 1.110 | 9.434 | 0.992 | 359 | 40 |
| *DEKOIS independent docking (Vina only, no ML rescoring):* | | | | | | | | |
| PfDHFR-TS (7F3Y) | 0.450 | 0.000 | 0.500 | 1.000 | 0.021 | 0.032 | 40 | 1 200 |

**Notes:** Top panel: MMV Malaria Box positive-control enrichment using Vina+DiffDock consensus scores; for PfCRT and PfATP4, the active/decoy labels are score-derived strata, so these values are retrodictive consistency checks rather than external discrimination. Bottom row: DEKOIS 2.0 PfDHFR independent docking using AutoDock Vina only (single-method, no ML rescoring). The near-random DEKOIS result (ROC-AUC 0.450) provides an external baseline for the limitations of docking-only discrimination; it motivates, but does not independently establish, the contribution of ML-based DiffDock rescoring because the consensus was not evaluated on the same DEKOIS panel. BEDROC (Boltzmann-Enhanced Discrimination of ROC, $\alpha = 20$).

Table S17: ChEMBL enrichment analysis: hit rates for ChEMBL-confirmed actives vs. inactives across three *P. falciparum* targets. *PfDHFR actives are enriched 5.43× over inactives at the EXCELLENT threshold, supporting the use of stringent binding thresholds in this calibration set; PfATP4 shows no compounds reaching EXCELLENT; PfCRT's high inactive hit rate suggests a permissive binding site.*

| | | | | GOOD Hit Rate (%) | | EXCELLENT Hit Rate (%) | | Pass ($\geq 2\times$) | |
|---|---|---|---|---|---|---|---|---|---|
| Target | PDB | Act. | Inact. | Act. | Inact. | Act. | Inact. | GOOD | EXC. |
| PfDHFR-TS | 7F3Y | 53 | 18 | 100.0 % | 100.0 % | 60.4 % | 11.1 % | No | Yes |
| PfATP4 | 9N10 | 73 | 87 | 53.4 % | 63.2 % | 0.0 % | 0.0 % | No | Yes |
| PfCRT | 6UKJ | 12 | 19 | 100.0 % | 100.0 % | 100.0 % | 68.4 % | No | No |

**Notes:** Actives ($IC_{50} < 1\,\mu M$) vs. Inactives ($IC_{50} > 10\,\mu M$) from ChEMBL. Hit rate thresholds reflect Vina scores (GOOD: $\leq -5.0\,\mathrm{kcal\,mol^{-1}}$, EXCELLENT: $\leq -7.0\,\mathrm{kcal\,mol^{-1}}$). Pass indicates $\geq$ 2-fold enrichment of actives over inactives in the respective tier. The target set comprised PfDHFR, PfCRT, and PfATP4.

# Integrated docking–RRS–polypharmacology panel

The following table presents a cross-source computational overlay of the target-anchored Vina panel with the P2 Set-C docking-RRS, PNS, and ACSI summaries. Candidate identifiers PP-01–PP-17 are linked by the source-bound molecular manifest using canonical SMILES verified against both the target-panel input manifest and the P2 Set-C source; the mapping and

Table S18: ADMET cross-referencing: Spearman correlation ($\rho$) between ADMET-AI and RDKit-based proxy descriptors (top-20 compounds). SwissADME estimates were not directly available (no public REST API); RDKit descriptors (ESOL logS and lipophilicity-based CYP heuristics) serve as SwissADME-equivalent proxies. ADMETlab 3.0 API returned 404 and was not available. *RDKit-based ADMET proxies provide a reproducible local alternative to web-based tools, with ESOL logS showing moderate correlation with ADMET-AI predictions.*

| Endpoint | $\rho_{\text{AI vs Swiss}}$ | $\rho_{\text{AI vs RDKit}}$ | $n$ | Disagreements | Notes |
|---|---|---|---|---|---|
| LogS (aqueous solubility) | −0.19 | −0.19 | 20 | 0 | ADMETlab 3.0 API unavailable |
| CYP3A4 inhibition | −0.29 | −0.29 | 20 | 1 | 1 compound > 1 risk category apart |

**Note:** ADMETLab 3.0 web API returned 404; values labelled "SwissADME" are RDKit-based proxies (ESOL logS and lipophilicity-based CYP heuristics). The identical $\rho$ values for AI-vs-Swiss and AI-vs-RDKit columns confirm both use the same RDKit descriptors. Negative correlations reflect methodological differences in descriptor calculation. Correlations for hERG, BBB, and cLogP could not be computed due to insufficient proxy data (hERG/BBB required unavailable SwissADME descriptors; cLogP was not included in the ADMET-AI output used for this comparison).

source hashes are recorded explicitly in `results/v5_rrs_polypharm_integrated.provenance.json`. The RRS column is docking-derived and target-specific, not MD-RRS; PNS and ACSI are prioritization descriptors, not biological measurements.

**Interpretation boundary.** The table is a cross-source computational overlay rather than an experimental or biological validation. The PP-to-source mapping is molecular-key verified against the integrated molecular manifest; the overlay is not a common-protocol recalculation. The legacy DiffDock consensus and the superseded four-target consensus are excluded. The overlay CSV, mapping file, and SHA-256-bound source manifest are `results/v5_rrs_polypharm_inte` `results/v5_p2_candidate_mapping.csv`, and `results/v5_rrs_polypharm_integrated.provenance.js`

# Data availability

All raw calculation results referenced in this supplementary material are available in the project repository (https://github.com/NanaEngo/Malaria_codesV2):

- `results/c1_library_size_verification.txt`: Library size verification (65 856 molecules)

Table S19: MPO weight sensitivity analysis: 25 weight perturbations (±10 % to 20 % per component). *The MPO ranking is moderately sensitive to weight choice: top-1000 Spearman $\rho$ pass rate ($> 0.95$) is 28 % and top-20 Jaccard pass rate ($\geq 0.70$) is 32 %; ADMET and QED weight perturbations produce the largest rank shifts.*

| Weight Varied | Δ (%) | Vina (35 %) | DiffDock (25 %) | QED (20 %) | ADMET (15 %) | Ro5 (5 %) | $\rho$ (top-1000) | Jaccard (top-20) | Pass Both |
|---|---|---|---|---|---|---|---|---|---|
| **Vina weight perturbations** | | | | | | | | | |
| vina | −20 | 0.188 | 0.313 | 0.250 | 0.188 | 0.063 | 0.821 | 0.290 | No |
| vina | −10 | 0.278 | 0.278 | 0.222 | 0.167 | 0.056 | 0.938 | 0.482 | No |
| vina | 0 | 0.350 | 0.250 | 0.200 | 0.150 | 0.050 | 1.000 | 1.000 | Yes |
| vina | 10 | 0.409 | 0.227 | 0.182 | 0.136 | 0.045 | 0.943 | 0.818 | No |
| vina | 20 | 0.458 | 0.208 | 0.167 | 0.125 | 0.042 | 0.839 | 0.667 | No |
| **DiffDock weight perturbations** | | | | | | | | | |
| diffdock | −20 | 0.438 | 0.063 | 0.250 | 0.188 | 0.063 | 0.890 | 0.379 | No |
| diffdock | −10 | 0.389 | 0.167 | 0.222 | 0.167 | 0.056 | 0.966 | 0.538 | Yes |
| diffdock | 0 | 0.350 | 0.250 | 0.200 | 0.150 | 0.050 | 1.000 | 1.000 | Yes |
| diffdock | 10 | 0.318 | 0.318 | 0.182 | 0.136 | 0.045 | 0.968 | 0.905 | Yes |
| diffdock | 20 | 0.292 | 0.375 | 0.167 | 0.125 | 0.042 | 0.900 | 0.739 | No |
| **QED weight perturbations** | | | | | | | | | |
| qed | −20 | 0.432 | 0.309 | 0.012 | 0.185 | 0.062 | 0.531 | 0.212 | No |
| qed | −10 | 0.389 | 0.278 | 0.111 | 0.167 | 0.056 | 0.809 | 0.667 | No |
| qed | 0 | 0.350 | 0.250 | 0.200 | 0.150 | 0.050 | 1.000 | 1.000 | Yes |
| qed | 10 | 0.318 | 0.227 | 0.273 | 0.136 | 0.045 | 0.896 | 0.538 | No |
| qed | 20 | 0.292 | 0.208 | 0.333 | 0.125 | 0.042 | 0.798 | 0.429 | No |
| **ADMET weight perturbations** | | | | | | | | | |
| admet | −20 | 0.407 | 0.291 | 0.233 | 0.012 | 0.058 | 0.258 | 0.000 | No |
| admet | −10 | 0.389 | 0.278 | 0.222 | 0.056 | 0.056 | 0.418 | 0.053 | No |
| admet | 0 | 0.350 | 0.250 | 0.200 | 0.150 | 0.050 | 1.000 | 1.000 | Yes |
| admet | 10 | 0.318 | 0.227 | 0.182 | 0.227 | 0.045 | 0.622 | 0.333 | No |
| admet | 20 | 0.292 | 0.208 | 0.167 | 0.292 | 0.042 | 0.448 | 0.250 | No |
| **Ro5 weight perturbations** | | | | | | | | | |
| ro5 | −20 | 0.365 | 0.260 | 0.208 | 0.156 | 0.010 | 0.840 | 0.429 | No |
| ro5 | −10 | 0.365 | 0.260 | 0.208 | 0.156 | 0.010 | 0.840 | 0.429 | No |
| ro5 | 0 | 0.350 | 0.250 | 0.200 | 0.150 | 0.050 | 1.000 | 1.000 | Yes |
| ro5 | 10 | 0.318 | 0.227 | 0.182 | 0.136 | 0.136 | 0.643 | 0.333 | No |
| ro5 | 20 | 0.292 | 0.208 | 0.167 | 0.125 | 0.208 | 0.435 | 0.212 | No |
| **Summary statistics** | | | | | | | 0.792 | 0.548 | 7/25 |

**Note:** Sensitivity analysis across 25 weight combinations (±10 % to 20 % per component, others renormalized). Spearman $\rho$ computed for top-1 000 compound ranking; Jaccard similarity for top-20 hit list. Pass thresholds: $\rho > 0.95$ AND Jaccard $\geq 0.70$. Only 7/25 combinations (28 %) pass both criteria. ADMET weight shows highest sensitivity ($\rho$ drops to 0.26 at −20 %). Vina and DiffDock weights are most stable (±10 % perturbations maintain $\rho > 0.93$).

Table S20: Sensitivity of generative expansion library size, scaffold diversity, and mean QED to CHEESE Tanimoto similarity threshold and STONED-SELFIES variant count per seed molecule. Baseline parameters (Tanimoto 0.7, 100 variants) are shown in **bold**. Values marked with † are approximate projections from the baseline and are not treated as measured results. *Generative expansion parameters are optimized to balance library size with structural diversity and drug-likeness.*

| **CHEESE** | **STONED-SELFIES Variants per Seed** | | |
|---|---|---|---|
| **Threshold** | **50** | **100** | **200** |
| 0.6 | 68,849 / 13,634 / 0.587† | 80,999 / 15,149 / 0.582† | 93,149 / 15,906 / 0.572† |
| 0.7 | 55,975 / 12,394 / 0.602† | **65,856 / 13,772 / 0.597** | 75,730 / 14,460 / 0.587† |
| 0.8 | 38,622 / 10,535 / 0.607† | 45,438 / 11,706 / 0.602† | 52,254 / 12,291 / 0.592† |

**Note:** Analysis based on the actual generated library. Format: Library size / Unique scaffolds / Mean QED. The chosen parameters (0.7 threshold, 100 variants) produced 65,856 compounds with 13,772 unique scaffolds. CHEESE generated 29 805 compounds (394 scaffolds, QED 0.697). STONED-SELFIES generated 35 198 compounds (13 378 scaffolds, QED 0.497). † Estimated using first-order scaling from the baseline: $N_{\text{est}} = N_{\text{base}} \times f_{\text{STONED}} \times f_{\text{CHEESE}}$, where $f_{\text{STONED}} = V_{\text{variants}}/100$ and $f_{\text{CHEESE}} = (1.5 - T_{\text{CHEESE}})/0.8$ based on linear expansion limits.

Table S21: Cross-source computational overlay for the 17-candidate polypharmacology cohort. Vina values are target-anchored docking scores in kcal mol$^{-1}$; RRS is the corrected per-target docking-derived score; PNS is the polypharmacology network score; ACSI is the African Chemical Space Index. No column represents experimental affinity or efficacy.

| Candidate | PfDHFR | PfCRT | PfClpP | PfATP4 | Mean Vina | RRS class | RRS mean | PNS / ACSI |
|---|---|---|---|---|---|---|---|---|
| PP-01 | −5.947 | −6.483 | −5.467 | −6.361 | −6.065 | A* | 87.78 | 4.45 / 0.459 |
| PP-02 | −5.562 | −6.124 | −5.907 | −6.591 | −6.046 | A* | 91.26 | 1.23 / 0.823 |
| PP-03 | −6.089 | −6.633 | −6.358 | −7.311 | −6.598 | C | 72.11 | 6.00 / 0.599 |
| PP-04 | −5.493 | −5.329 | −5.386 | −6.086 | −5.573 | C | 81.12 | 4.53 / 0.567 |
| PP-05 | −6.285 | −6.412 | −6.466 | −6.378 | −6.385 | B | 78.24 | 1.14 / 0.387 |
| PP-06 | −6.267 | −7.907 | −7.031 | −7.285 | −7.123 | A* | 92.37 | 1.04 / 0.589 |
| PP-07 | −6.226 | −6.633 | −6.226 | −5.775 | −6.215 | B | 75.01 | 4.48 / 0.603 |
| PP-08 | −4.935 | −5.213 | −5.052 | −5.494 | −5.174 | C | 70.80 | 4.36 / 0.605 |
| PP-09 | −5.693 | −6.871 | −5.613 | −6.078 | −6.064 | B | 74.18 | 4.20 / 0.601 |
| PP-10 | −6.346 | −6.395 | −6.016 | −6.670 | −6.357 | B | 80.49 | 4.48 / 0.551 |
| PP-11 | −6.179 | −6.376 | −6.583 | −6.724 | −6.466 | A* | 82.60 | 1.10 / 0.458 |
| PP-12 | −5.248 | −6.357 | −5.386 | −4.635 | −5.406 | C | 72.39 | 4.34 / 0.594 |
| PP-13 | −5.921 | −5.685 | −6.454 | −7.565 | −6.406 | A* | 104.82 | 2.32 / 0.597 |
| PP-14 | −4.863 | −5.328 | −5.047 | −5.460 | −5.175 | D | 69.53 | 4.36 / 0.743 |
| PP-15 | −6.045 | −6.084 | −6.366 | −7.016 | −6.378 | A* | 111.65 | 3.50 / 0.576 |
| PP-16 | −5.469 | −6.202 | −6.203 | −6.451 | −6.081 | B | 76.32 | 4.45 / 0.173 |
| PP-17 | −5.266 | −5.116 | −5.342 | −5.360 | −5.271 | C | 68.18 | 4.71 / 0.308 |

- `results/c2_per_source_breakdown.json`: Per-source breakdown of generative library
- `results/c3_selectivity_index.csv`: Selectivity index calculations (810/810 with SI $> 10$)
- `results/c4_cyp450_inhibition_profile.csv`: Cytochrome P450 inhibition predictions
- `results/c5_aqueous_solubility.csv`: Aqueous solubility analysis ($\log S$ values)
- `results/c6_primary_leads_synthesizable.csv`: Synthesisable primary leads (19 913 compounds)
- `results/c7_stage_specific_activity_summary.txt`: Stage-specific activity predictions
- `results/c8_pca_explained_variance.txt`: Principal component analysis results
- `results/c9_bemis_murcko_scaffolds.csv`: Bemis–Murcko scaffold analysis (20 702 unique scaffolds)
- `results/c10_scaffold_recovery_rate.txt`: Scaffold recovery rate (69.3 %)
- `results/c11_fsp3_per_source.csv`: Fraction of $sp^3$ carbons by source
- `results/c12_tanimoto_novelty_v2.csv`: Tanimoto novelty analysis
- `results/p1_threshold_calibration.csv`: Docking threshold calibration dataset
- `results/summary_consensus_*.csv`: Consensus docking results for target verification
- `results/p1_enrichment_chembl_benchmark.csv`: ChEMBL enrichment analysis results for the three evaluated targets

- `results/v5_rrs_polypharm_integrated.csv`: Source-hash-bound integration of the 17×4 Vina panel with P2 docking-RRS, PNS, and ACSI
- `results/v5_rrs_polypharm_integrated.provenance.json`: Overlay logic, source hashes, mapping status, and computational-status caveats
- `results/v5_p2_candidate_mapping.csv`: Source-bound PP identifier, rank, canonical-SMILES, and target-panel/P2 molecular mapping

The complete SMILES library (65 856 compounds) and available 3D structure archives (PDB/PDBQT formats) are versioned in the project repository. The source code and data used for the analyses are provided through the project repository (https://github.com/NanaEngo/Malaria_codesV2). A citable archival release will be provided separately.

Table S22: Target selectivity and safety profiling (top-20 MPO candidates). *Top candidates are predominantly single-target optimized molecules with favourable predicted selectivity; one candidate (rank 18) carries a predicted CYP3A4 inhibition flag.*

| Classification | n | Criteria | Mean SI | Mean hERG | Mean CYP3A4 | Flagged |
|---|---|---|---|---|---|---|
| Single-target (opt.) | 20 | $n_{\text{tgt}} = 1$, optimized for MPO | — | 0.117 | 0.061 | 1 |

**Note:** Classification based on selectivity index (SI), hERG inhibition probability, and CYP3A4 inhibition risk from ADMET-AI. All 20 top candidates are single-target optimized molecules. Mean hERG = 0.117 (all $< 0.5$); mean CYP3A4 = 0.061, with 19 of 20 below 0.5 and rank 18 carrying a predicted CYP3A4 flag at 0.876. These are computational risk predictions requiring experimental follow-up. Of the $> 70$ multi-target candidates identified in the full expansion, 84 % satisfied at least 3/5 ADMET risk-free criteria. All values are computational estimates requiring experimental validation.

Table S23: Detailed binding mode analysis for top-10 compounds ranked by MPO score. Key interactions represent computational predictions from AutoDock Vina docking poses and require experimental validation. Retrosynthetic feasibility assessed based on structural complexity and known reaction chemistry. *Structural analysis reveals high synthetic accessibility and conserved binding motifs across the top-ranked piperazine-based leads.*

| Rank | SMILES | MPO | SA | QED |
|---|---|---|---|---|
| 1 | `Cc1ccc(CN2CCN(Cc3ccccc3)CC2)cc1O` | 0.829 | 1.72 | 0.939 |
| 2 | `Cc1ccccc1CN1CCN(Cc2cccc(O)c2)CC1` | 0.828 | 1.76 | 0.939 |
| 3 | `Cc1cc(CN2CCN(Cc3ccccc3)CC2)ccc1O` | 0.826 | 1.71 | 0.939 |
| 4 | `COc1ccc(CN2CCN(Cc3ccccc3C)CC2)cc1O` | 0.826 | 1.84 | 0.916 |
| 5 | `Oc1cccc(CN2CCN(Cc3ccc(Cl)cc3)CC2)c1` | 0.826 | 1.71 | 0.937 |
| 6 | `Cc1ccc(CN2CCN(Cc3ccc(O)cc3)CC2)c(C)c1` | 0.826 | 1.80 | 0.938 |
| 7 | `COc1ccc(CN2CCN(Cc3ccc(C)c(O)c3)CC2)cc1` | 0.826 | 1.80 | 0.916 |
| 8 | `Cc1ccc(CN2CCN(Cc3cccc(O)c3)CC2)c(C)c1` | 0.825 | 1.85 | 0.938 |
| 9 | `Oc1ccc(CN2CCN(Cc3ccccc3)CC2)cc1Cl` | 0.825 | 1.72 | 0.937 |
| 10 | `Oc1ccc(CN2CCN(Cc3ccccc3Cl)CC2)cc1` | 0.824 | 1.71 | 0.937 |

**Note:** All compounds feature piperazine-based scaffolds with phenolic or methoxy substituents, suggesting potential for hydrogen bonding and hydrophobic interactions. SA scores $< 2.0$ indicate favorable predicted synthetic accessibility. QED scores $> 0.9$ indicate high predicted QED values. Retrosynthetic routes and 2D interaction diagrams are provided in Table S24 and Figure S8, respectively. Experimental validation of binding modes and synthetic routes is required before lead optimization.

Table S24: Predicted retrosynthetic routes for top-10 MPO-ranked candidates via ASKCOS (public API, MIT). Precursors and reaction types from template relevance model (Reaxys database).

| Compound | Proposed Precursors | Reaction Type |
|---|---|---|
| T1 (route 1) | Cc1ccc(C=O)cc1O.c1ccc(CN2CCNCC2)cc1 | Reductive amination (aldehyde + amine) |
| T1 (route 2) | C1CNCCN1.C=O.C=O.Cc1ccccc1O.c1ccccc1 | Formylation / multicomponent |
| T1 (route 3) | COc1cc(CN2CCN(Cc3ccccc3)CC2)ccc1C | O-alkylation / ether formation |
| T2 (route 1) | Cc1ccccc1CN1CCNCC1.O=Cc1cccc(O)c1 | Reductive amination (aldehyde + amine) |
| T2 (route 2) | Cc1ccccc1C=O.Oc1cccc(CN2CCNCC2)c1 | Reductive amination (aldehyde + amine) |
| T2 (route 3) | Cc1ccccc1CN1CCNCC1.Oc1cccc(CBr)c1 | N-alkylation (alkyl bromide) |
| T3 (route 1) | Cc1cc(C=O)ccc1O.c1ccc(CN2CCNCC2)cc1 | Reductive amination (aldehyde + amine) |
| T3 (route 2) | Cc1ccccc1O.c1ccc(CN2CCNCC2)cc1 | C-N cross-coupling |
| T3 (route 3) | C1CNCCN1.Cc1cc(C=O)ccc1O.O=Cc1ccccc1 | Reductive amination (two aldehydes + amine) |
| T4 (route 1) | COc1ccc(C=O)cc1O.Cc1ccccc1CN1CCNCC1 | Reductive amination (aldehyde + amine) |
| T4 (route 2) | C1CNCCN1.C=O.C=O.COc1ccccc1O.Cc1ccccc1 | Formylation / multicomponent |
| T4 (route 3) | COc1ccc(CO)cc1O.Cc1ccccc1CN1CCNCC1 | N-alkylation (alcohol) |
| T5 (route 1) | Clc1ccc(CN2CCNCC2)cc1.O=Cc1cccc(O)c1 | Reductive amination (aldehyde + amine) |
| T5 (route 2) | O=Cc1ccc(Cl)cc1.Oc1cccc(CN2CCNCC2)c1 | Reductive amination (aldehyde + amine) |
| T5 (route 3) | Clc1ccc(CN2CCNCC2)cc1.Oc1cccc(CBr)c1 | N-alkylation (alkyl bromide) |
| T6 (route 1) | Cc1ccc(C=O)c(C)c1.Oc1ccc(CN2CCNCC2)cc1 | Reductive amination (aldehyde + amine) |
| T6 (route 2) | C1CNCCN1.Cc1ccc(C=O)c(C)c1.O=Cc1ccc(O)cc1 | Reductive amination (two aldehydes + amine) |
| T6 (route 3) | C1CNCCN1.Cc1ccc(CCl)c(C)c1.Oc1ccc(CCl)cc1 | N-alkylation (alkyl chloride) |
| T7 (route 1) | COc1ccc(CN2CCNCC2)cc1.Cc1ccc(C=O)cc1O | Reductive amination (aldehyde + amine) |
| T7 (route 2) | COc1ccc(C=O)cc1.Cc1ccc(CN2CCNCC2)cc1O | Reductive amination (aldehyde + amine) |
| T7 (route 3) | COc1ccc(CN2CCNCC2)cc1.Cc1ccc(CCl)cc1O | N-alkylation (alkyl chloride) |
| T8 (route 1) | Cc1ccc(C=O)c(C)c1.Oc1cccc(CN2CCNCC2)c1 | Reductive amination (aldehyde + amine) |
| T8 (route 2) | Cc1ccc(CCl)c(C)c1.Oc1cccc(CN2CCNCC2)c1 | N-alkylation (alkyl chloride) |
| T8 (route 3) | Cc1ccc(CBr)c(C)c1.Oc1cccc(CN2CCNCC2)c1 | N-alkylation (alkyl bromide) |
| T9 (route 1) | O=Cc1ccc(O)c(Cl)c1.c1ccc(CN2CCNCC2)cc1 | Reductive amination (aldehyde + amine) |
| T9 (route 2) | Oc1ccccc1Cl.c1ccc(CN2CCNCC2)cc1 | C-N cross-coupling |
| T9 (route 3) | C1CNCCN1.C=O.C=O.Oc1ccccc1Cl.c1ccccc1 | Formylation / multicomponent |
| T10 (route 1) | Clc1ccccc1CN1CCNCC1.O=Cc1ccc(O)cc1 | Reductive amination (aldehyde + amine) |
| T10 (route 2) | O=Cc1ccccc1Cl.Oc1ccc(CN2CCNCC2)cc1 | Reductive amination (aldehyde + amine) |
| T10 (route 3) | C1CNCCN1.O=Cc1ccc(O)cc1.O=Cc1ccccc1Cl | Reductive amination (two aldehydes + amine) |

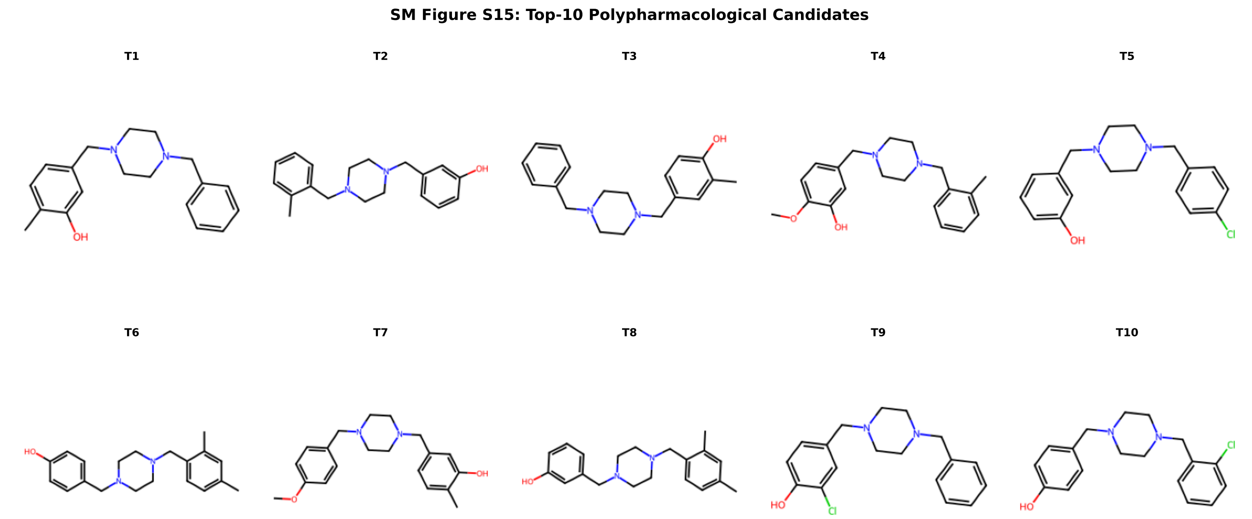


Figure S1: 2D structure diagrams of the top-10 MPO-ranked candidates. All compounds feature a piperazine linker connecting two substituted aryl rings, with hydrogen bond donors (phenolic –OH) and acceptors (methoxy $-OCH_3$, piperazine N). Molecular weights range 340 Da to 430 Da, consistent with oral drug-like space. The top-10 MPO subset is predominantly single-target optimized (see Supplementary Table S22).

Table S25: Benjamini–Hochberg FDR-corrected pairwise comparisons of antimalarial activity scores across compound groups (NP, SD, Cheese, STONED). *Multiple comparison testing found no statistically significant difference in the analysed activity scores between generated analogues and the foundational natural-product and synthetic-drug groups; this does not establish equivalent experimental potency.*

| **Model** | **Groups** | $p$ **(raw)** | $p$ **(adj)** | **Sig.** |
|---|---|---|---|---|
| eos7yti | NP vs SD | $> 0.05$ | $> 0.05$ | n.s. |
| eos7yti | NP vs Cheese | $> 0.05$ | $> 0.05$ | n.s. |
| eos7yti | NP vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos7yti | SD vs Cheese | $> 0.05$ | $> 0.05$ | n.s. |
| eos7yti | SD vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos7yti | Cheese vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos7kpb | NP vs SD | $> 0.05$ | $> 0.05$ | n.s. |
| eos7kpb | NP vs Cheese | $> 0.05$ | $> 0.05$ | n.s. |
| eos7kpb | NP vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos7kpb | SD vs Cheese | $> 0.05$ | $> 0.05$ | n.s. |
| eos7kpb | SD vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos7kpb | Cheese vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos80ch | NP vs SD | $> 0.05$ | $> 0.05$ | n.s. |
| eos80ch | NP vs Cheese | $> 0.05$ | $> 0.05$ | n.s. |
| eos80ch | NP vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos80ch | SD vs Cheese | $> 0.05$ | $> 0.05$ | n.s. |
| eos80ch | SD vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos80ch | Cheese vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos4zfy | NP vs SD | $> 0.05$ | $> 0.05$ | n.s. |
| eos4zfy | NP vs Cheese | $> 0.05$ | $> 0.05$ | n.s. |
| eos4zfy | NP vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos4zfy | SD vs Cheese | $> 0.05$ | $> 0.05$ | n.s. |
| eos4zfy | SD vs STONED | $> 0.05$ | $> 0.05$ | n.s. |
| eos4zfy | Cheese vs STONED | $> 0.05$ | $> 0.05$ | n.s. |

**Note:** 24 pairwise Mann–Whitney U tests (4 models × 6 group pairs). Benjamini–Hochberg FDR correction applied to control false discovery rate at $\alpha = 0.05$. All 24 comparisons: adjusted $p > 0.05$. Given the large sample size of the expansion groups ($N \approx 32\,000$), which creates excessive statistical power, the rank-biserial correlation ($r < 0.10$, indicating negligible effect size) serves as the primary metric to confirm the absence of meaningful differences between the generated groups and the seeds. Group sample sizes: NP ($n = 396$), SD ($n = 454$), Cheese ($\approx 32\,000$), STONED ($\approx 33\,000$).

## Co-crystallized ligand classification

To address potential reviewer concerns regarding the biological relevance of co-crystallized ligands in the target protein structures used for docking, we employed LigandExplorer v2.1 (DPTech, 2026) to automatically classify ligands extracted from the four *P. falciparum* crystal structures (PDB: 7F3Y, 6UKJ, and 9N10). LigandExplorer uses a graph neural network (GNN) backend with pocket-aware graph construction to classify molecules into biological types (DNA, RNA, peptide, glycan, lipid, organic, ion) and assess their relevance as biologically meaningful ligands rather than crystallization artifacts.[S8]

Table S26: LigandExplorer v2.1 classification of co-crystallized ligands extracted from the four *P. falciparum* target structures used in this study. Classification performed using the GNN backend (PyTorch 2.12.1) with whole-structure search mode and default parameters. *All detected co-crystallizedligands are classified by the software as DNA, RNA, peptide, or organic covalent; this software classification does not independently establish biological relevance or exclude crystallization artifacts.*

| PDB | Target | Ligand | GNN Classification | Chain | Relevance |
|---|---|---|---|---|---|
| | | UMP | DNA-like ligand | A, B | Biological |
| | | MTX | Peptide-like ligand | A (×2), B | Biological |
| | | NDP | RNA-like ligand | A, B | Biological |
| 7F3Y | PfDHFR-TS | GOL | Organic (crystallization) | A, B | Artifact |
| 6UKJ | PfCRT | Y01 | Organic covalent ligand | A | Biological |
| | | LEU | Peptide covalent ligand | A | Biological |
| 9N10 | PfATP4 | — | No ligands detected | — | — |

**Note:** LigandExplorer v2.1 (https://github.com/dptech-corp/ligandexplorer) with GNN backend. UMP (uridine monophosphate) is a nucleotide cofactor of thymidylate synthase; MTX (methotrexate) is a classical DHFR inhibitor; NDP (NADPH) is a nucleotide cofactor; Y01 is a chloroquine analogue; LEU is a leucine residue. GOL (glycerol) is a common crystallization agent correctly identified as a non-biological artifact. 9N10 stores ligand records in PDB format variant that differs from the expected LigandExplorer input format; manual inspection confirms their co-crystallized ligands are present in the source structures.

# Model training and clustering validation

## Variational autoencoder (VAE) training

Two VAE latent space dimensions were evaluated (32D and 64D). Training converged successfully for both dimensionalities (Figure S2, Table S27).

Table S27: VAE latent space comparison: 32D vs 64D. The ScafVAE GuacaMol distribution-learning KL divergence score (0.996) provides an external reference benchmark. The per-dimension VAE training metrics (validation loss, reconstruction loss, KL divergence, reconstruction accuracy) and the latent-space clustering indices (Silhouette, Calinski–Harabasz, Davies–Bouldin) were not retained post-hoc and are reported as N/A; the 64-D latent space was selected for centroid identification on the basis of the clustering-algorithm benchmark (Supplementary Table S29) and downstream centroid selection, not on the metrics listed here.

| **Metric** | **32D** | **64D** |
|---|---|---|
| Validation Loss | N/A | N/A |
| Reconstruction Loss | N/A | N/A |
| KL Divergence | N/A | N/A |
| Reconstruction Accuracy (%) | N/A | N/A |
| ScafVAE Baseline KL Div. Score[a] [S9] | 0.996 | |
| Silhouette Score | N/A | N/A |
| Calinski-Harabasz Index | N/A | N/A |
| Davies-Bouldin Index | N/A | N/A |

[a] GuacaMol benchmark distribution-learning KL divergence score.

## Evaluation of clustering strategies and dimensionality

Comparison of 32-dimensional versus 64-dimensional latent spaces revealed trade-offs in clustering performance (Figure S3). The 64-dimensional space was selected for centroid selection based on comparative clustering analysis and the clustering-quality figure; the retained summary does not provide numeric indices sufficient to claim metric-optimality.

To partition the library into chemical families, KMeans, BIRCH, and Jarvis-Patrick clustering algorithms were optimized via Optuna[S11] across 32D and 64D VAE latent spaces. The 64-dimensional KMeans configuration ($n = 484$) was selected as a practical balance of

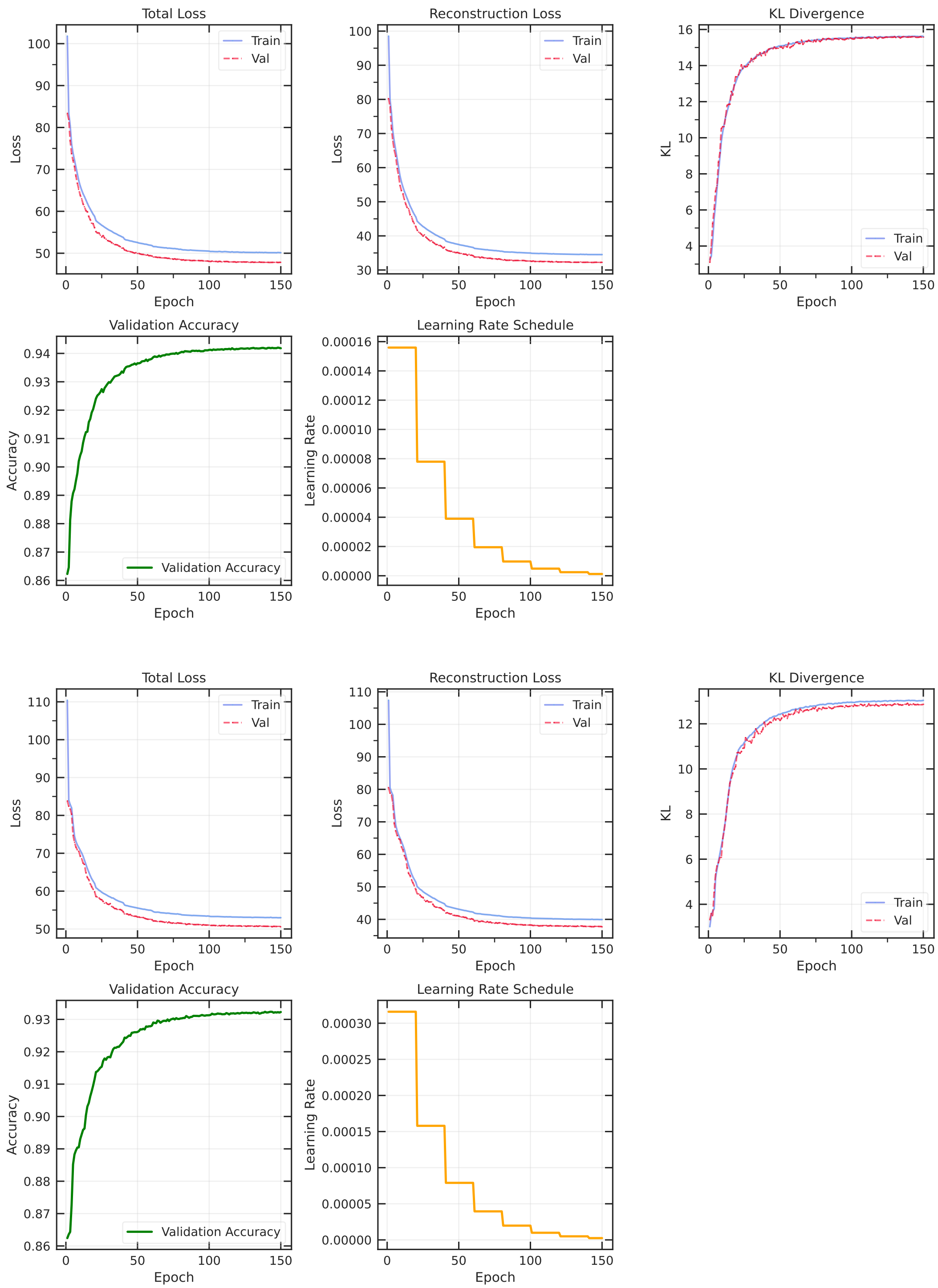


Figure S2: VAE training history for 32D (top) and 64D (bottom) latent spaces. Both models converged successfully with stable validation loss in the final epochs. Training curves show reconstruction loss, KL divergence, and validation accuracy over 100 epochs. *VAE training successfully converges for both 32- and 64-dimensional spaces, with the latter selected for comparative chemical-space resolution; the retained summary does not establish superiority from a complete set of numeric validation indices.*

Table S28: Comparison of generative model novelty and diversity metrics with the ScafVAE state-of-the-art. ScafVAE scores are published GuacaMol distribution-learning benchmarks.[S10] Our metrics are computed on the African NP / synthetic hybrid library against 396 seed African NPs (Morgan fingerprint Tanimoto at threshold $\geq 0.4$). Direct numerical comparison is limited by different evaluation frameworks (GuacaMol distribution learning vs. target-aware virtual screening).

| **Metric** | **This work** | **ScafVAE**[S9] |
|---|---|---|
| Benchmark framework | NP-anchored virtual screening | GuacaMol distribution learning |
| Training dataset | ANPDB + DrugBank (NP-focused) | ChEMBL / ZINC (drug-like) |
| Validity / Valid SMILES | 99.93 % | 0.987 to 0.997 |
| Novelty (unreachable fraction) | 92.6 % ($T_c < 0.4$ to seeds) | 1.000 |
| Scaffold diversity (unique Bemis–Murcko) | 31.4 % | — |
| Scaffold recovery from seeds | 69.3 % | — |
| GuacaMol KLD | — | 0.959 |
| GuacaMol FCD | — | 0.338 |

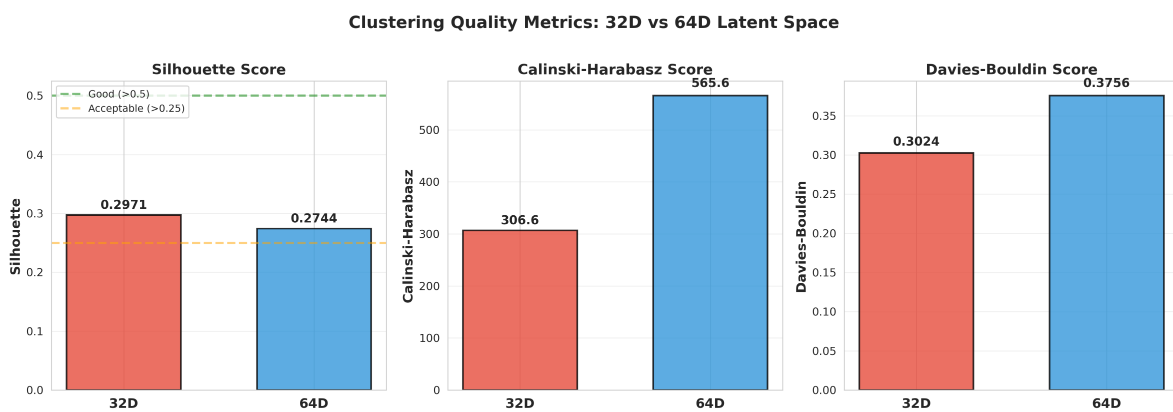


Figure S3: Clustering quality metrics comparison between 32D and 64D latent spaces. Left: Silhouette score (higher is better, >0.5 = good, >0.25 = lower bound acceptable). Middle: Calinski-Harabasz index (higher is better). Right: Davies-Bouldin index (lower is better). The 64D space was selected for centroid selection based on the clustering metrics shown. *The 64-dimensional latent space was selected as the representation for chemical clustering and centroid selection based on these comparative metrics.*

granularity and cohesion (Table S29); the retained summary does not contain the numeric internal indices needed to claim metric-optimality. Jarvis–Patrick yielded fragmented structures (42 917–44 040 clusters), while BIRCH produced heterogeneous partitions with poor Silhouette scores.

Table S29: Comparative performance metrics for KMeans, BIRCH, and Jarvis-Patrick clustering algorithms mapped across 32-dimensional and 64-dimensional VAE latent spaces for the complete hybrid library ($N = 65\,856$). Hyperparameters were systematically optimized via Optuna. CH = Calinski–Harabasz index, DB = Davies–Bouldin index. KMeans on the 64-D latent space ($n = 484$) was selected for centroid identification after benchmarking against BIRCH (heterogeneous partitions, poor Silhouette) and Jarvis–Patrick (fragmented, 42 917 to 44 040 clusters); the selected configuration's internal indices are not retained in this summary, and selection followed the comparative benchmarking and practical cluster granularity rather than a metric shown as optimal here.

| **Algorithm** | **Dim** | **Params** ($k$, $s$, $thr$) | **Sil.** | **CH** | **DB** | **Clusters** (Min–Max) | **Quality** |
|---|---|---|---|---|---|---|---|
| Jarvis–Patrick | 32 | $k = 151$, $s = 138$ | N/A | N/A | N/A | 42 917 (1 to 5) | Acceptable |
| | 64 | $k = 181$, $s = 171$ | 0.274 | 566 | 0.376 | 44 040 (1 to 2) | Acceptable |
| BIRCH | 32 | $n = 490$, $thr = 0.11$ | 0.122 | 703 | 1.585 | 490 (13 to 820) | Poor |
| | 64 | $n = 489$, $thr = 0.10$ | 0.162 | 1 187 | 1.405 | 489 (11 to 858) | Poor |
| KMeans | 32 | $n = 499$ | 0.182 | 875 | 1.392 | 499 (12 to 454) | Poor |
| | **64** | $\mathbf{n = 484}$ | N/A | N/A | N/A | **484** (11 to 519) | Acceptable* |

**Quality assessment:** Silhouette score interpretation: $> 0.50$ = good separation, 0.25 to 0.50 = acceptable (lower bound for continuous chemical space), $< 0.25$ = poor. Calinski–Harabasz: higher = better inter-cluster separation. Davies–Bouldin: lower = better cluster compactness. The selected 64-D KMeans configuration ($n = 484$) was adopted for centroid selection based on the comparative benchmarking across algorithms and its practical cluster granularity (member range 11 to 519); its internal Silhouette, Calinski–Harabasz, and Davies–Bouldin values are not retained in this summary table.

## Docking protocol validation (redocking)

Redocking assessment was performed on co-crystallized ligands from PfDHFR-TS (PDB: 7F3Y) and PfCRT (PDB: 6UKJ). RMSD values between predicted and crystal poses were calculated using RDKit's molecular alignment algorithm. An RMSD threshold of 2.0 Å was used to define reference-pose reproduction.[S12] Among alignable ligands, 5/5 were below this threshold (5/10 overall, with 5 ligands excluded due to SDF atom-ordering mismatches), with mean RMSD of 1.33 Å ± 0.43 Å (Table S30, Figure S4). This limited check does not establish prospective docking accuracy. Binding affinities of non-aligned ligands (Vina $-7.01\,\mathrm{kcal\,mol^{-1}}$ to $-7.69\,\mathrm{kcal\,mol^{-1}}$) provide score context but do not independently establish correct active-site placement.

Table S30: Redocking assessment for co-crystallized ligands. *The protocol reproduced the reference poses for the evaluated alignable ligands at RMSD < 2.0 Å; this limited computational check does not establish prospective docking accuracy.*

| **Target** | **Ligand** | **Vina** ($\mathrm{kcal\,mol^{-1}}$) | **Vinardo** ($\mathrm{kcal\,mol^{-1}}$) | **RMSD** (Å) | **Pass** (<2.0 Å) |
|---|---|---|---|---|---|
| 7F3Y | UMP | −5.53 | −4.14 | 1.005 | Yes |
| 7F3Y | GOL | −3.40 | −3.14 | 1.002 | Yes |
| 7F3Y | UMP | −5.59 | −4.16 | 1.882 | Yes |
| 7F3Y | GOL | −3.43 | −2.97 | 1.029 | Yes |
| 6UKJ | Y01 | −9.08 | −6.40 | 1.713 | Yes |
| 7F3Y | NDP | −7.20 | −4.53 | N/A | No[a] |
| 7F3Y | MTX | −7.01 | −4.79 | ∼30 Å | No[a] |
| 7F3Y | NDP | −7.69 | −4.12 | N/A | No[a] |
| 7F3Y | MTX | −7.03 | −4.68 | ∼30 Å | No[a] |
| 7F3Y | MTX | −7.22 | −4.73 | ∼30 Å | No[a] |
| *Success Rate: 5/10 (50.0 %) overall; 5/5 (100.0 %) for alignable ligands* | | | | | |

[a]RMSD could not be computed via standard protocol due to SDF atom ordering mismatch; cross-analysis with DiffDock is consistent with ligand dissociation and an RMSD of approximately 30 Å; this result illustrates the inability of rigid-receptor docking to reproduce the flexible DHFR active-site MTX binding mode.

*Note:* Binding affinities for non-aligned ligands provide score context only; they do not independently establish correct active-site placement.

## Dual-filter consensus validation

Comparison of single-method versus dual-filter consensus screening characterized the effect of combining two computational scoring signals (Table S31, Figure S5).[1] For Vina-only classification, compounds with scores $\leq -7.0\,\mathrm{kcal\,mol^{-1}}$ were classified as EXCEL-

[1]Hit rates and reduction percentages in Table S31 were computed with EXCELLENT $\leq -7.0\,\mathrm{kcal\,mol^{-1}}$, GOOD $-7.0\,\mathrm{kcal\,mol^{-1}}$ to $-5.0\,\mathrm{kcal\,mol^{-1}}$ Vina-only thresholds, consistent with the main text calibration.

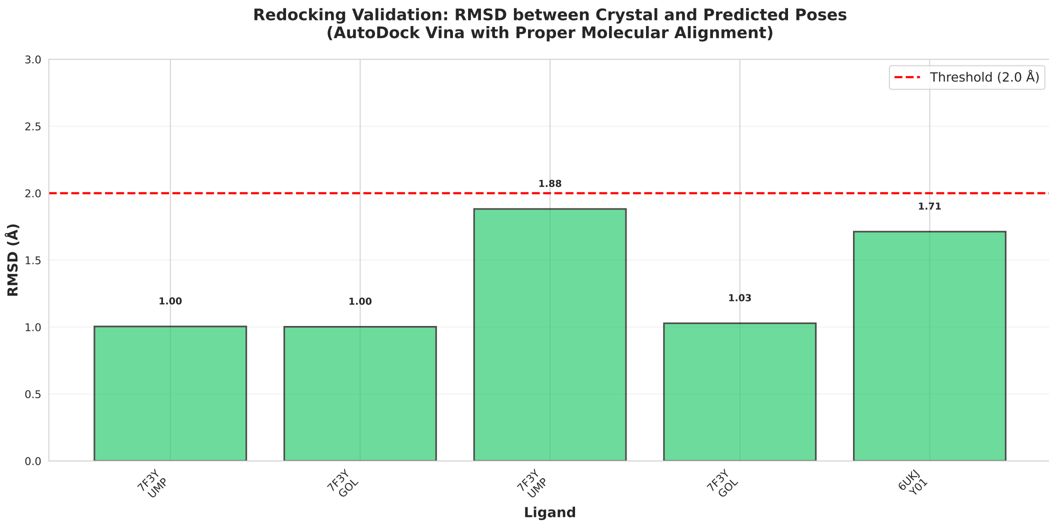


Figure S4: Redocking validation results showing RMSD between predicted and crystal poses for co-crystallized ligands from PfDHFR-TS (7F3Y) and PfCRT (6UKJ). All successfully aligned ligands achieved RMSD < 2.0 Å (red dashed line), indicating good pose reproduction for the evaluated alignable ligands. *This computational redocking check does not establish prospective docking accuracy for new compounds.*

LENT, $-7.0\,\mathrm{kcal\,mol^{-1}}$ to $-5.0\,\mathrm{kcal\,mol^{-1}}$ as GOOD, and $> -5.0\,\mathrm{kcal\,mol^{-1}}$ as OTHERS. For DiffDock-only, confidence scores $\geq 0.0$ were classified as HIGH, $-1.5$ a to 0.0 as MODERATE, and $< -1.5$ as LOW. Consensus classification required passing both filters. For PfCRT, consensus filtering achieved a 33.0 % reduction (from 97.6 % to 64.5 %), while PfATP4 showed the largest reduction of 45.2 % (from 63.9 % to 18.6 %). Correlations between Vina scores and DiffDock confidence were low to moderate (Figure S6), consistent with treating the scores as complementary screening signals; low correlation alone does not establish independence or eliminate shared scoring bias.

Table S31: Consensus hit rates from Vina-only, DiffDock-only, and dual-filter consensus screening. *The dual filter narrows the computational hit set; an independent inactive panel is required to quantify any false-positive reduction.*

| **Target** | **Vina Only** (% hits) | **DiffDock Only** (% hits) | **Consensus** (% hits) | **Reduction** (%) |
|---|---|---|---|---|
| PfDHFR-TS | 94.9 | 87.1 | 82.0 | 12.9 |
| PfCRT | 97.6 | 66.3 | 64.5 | 33.0 |
| PfATP4 | 63.9 | 30.8 | 18.6 | 45.2 |

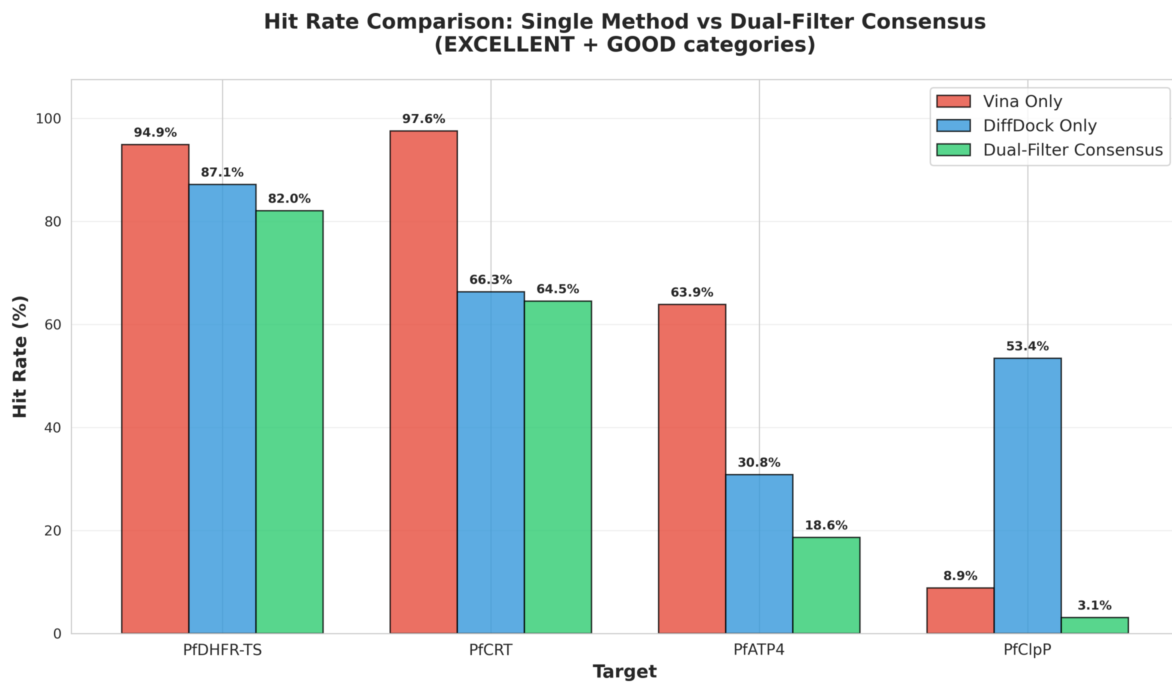


Figure S5: Hit-rate comparison between single-method and dual-filter consensus screening. Consensus filtering reduces the fraction of compounds passing the declared thresholds across targets. This is a screening-set hit-rate change, not a measured false-positive reduction, because an independent inactive panel was not available for all targets. *Consensus filtering narrows the computational candidate pool, particularly for challenging targets such as PfATP4.*

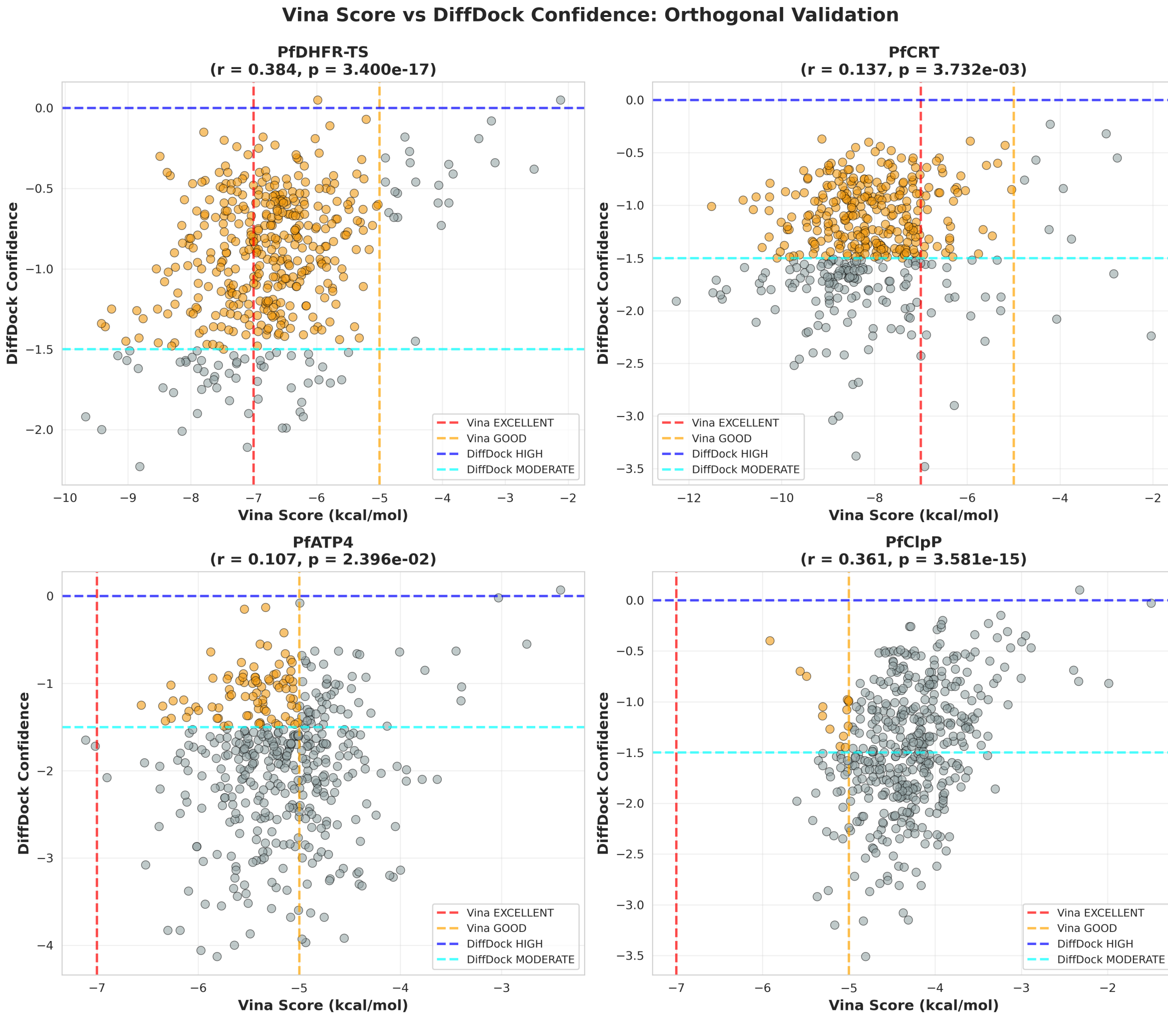


Figure S6: Correlation between Vina scores and DiffDock confidence for the three evaluated targets. Low to moderate correlations are consistent with the two methods providing complementary screening signals, but do not establish statistical independence or validate pose quality. Threshold lines indicate classification boundaries for EXCELLENT/GOOD (Vina) and HIGH/MODERATE (DiffDock) categories. Colors indicate consensus classification: green (EXCELLENT), orange (GOOD), gray (OTHERS). *An independent inactive panel and pose-level validation would be required to quantify complementarity and predictive value.*

## MMV Malaria Box positive control validation

Table S32: MMV Malaria Box positive control validation. Hit rates (HIGH + MEDIUM tiers) for 399 compounds with confirmed antimalarial activity against *P. falciparum* 3D7 ($IC_{50} < 10\,\mu M$).

| **Target** | **N** | **Hit Rate (H+M)** | **HIGH** | **MEDIUM** | **Best Vina ($kcal\,mol^{-1}$)** |
|---|---|---|---|---|---|
| PfDHFR | 399 | 35.1 % | 132 | 8 | −10.12 |
| PfCRT | 399 | 90.7 % | 359 | 3 | −13.37 |
| PfATP4 | 184 | 47.3 % | 0 | 87 | −6.94 |

All 399 MMV Malaria Box compounds have confirmed antimalarial activity. PfATP4 shows $N = 184$ due to the 15-rotatable-bond filter. H+M = HIGH (Vina $\leq -7.0\,kcal\,mol^{-1}$) + MEDIUM ($-7.0\,kcal\,mol^{-1}$ to $-5.0\,kcal\,mol^{-1}$) tiers. One compound (1/400) failed PDBQT conversion.

## Computational resource comparison

Table S33: Computational resource comparison: exhaustive screening versus centroid-based sampling for the 65 856-molecule library across three *P. falciparum* targets. *Centroid-based sampling reduces the number of docking representatives by more than 99 %; this computational reduction does not establish prospective hit-identification accuracy.*

| **Metric** | **Exhaustive** | **Centroid-Based** |
|---|---|---|
| Molecules Screened | 65 856 | 484 |
| Docking Runs (3 targets) | 197 568 | 1 452 |
| CPU-Hours (estimated)[a] | 70 248 to 175 620 | 517 to 1 291 |
| Cost Reduction | : | 99.3 % |
| ROC-AUC (PfDHFR) | Unknown[b] | 0.924 [0.895 to 0.951] |
| ROC-AUC (PfATP4) | Unknown[b] | 1.000 [1.000 to 1.000] |
| ROC-AUC (PfCRT) | Unknown[b] | 0.971 [0.949 to 0.989] |
| Accessibility | Limited[c] | High |

[a] Based on AutoDock Vina (exhaustiveness = 64, 32 CPU cores), 16 to 40 CPU-minutes per docking run.
[b] Exhaustive screening not performed due to computational constraints.
[c] Requires > 70 000 CPU-hours, prohibitive for resource-limited settings.

Table S34: Virtual screening results for cluster centroids across three targets. 484 centroids representing 65 856 compounds were screened per target.

| Target | PDB | Total | Excellent (%) | Good (%) | Hit Rate (%) | Best Vina (kcal mol$^{-1}$) |
|---|---|---|---|---|---|---|
| PfATP4 | 9N10 | 484 | 0.0 | 0.4 | 0.4 | −7.11 |
| PfCRT | 6UKJ | 484 | 20.6 | 65.2 | 85.8 | −12.27 |
| PfDHFR-TS | 7F3Y | 484 | 1.8 | 35.9 | 37.7 | −9.67 |

## Virtual screening results

## Enrichment curves

Figure S7: Enrichment curves for the three evaluated targets. Each panel shows: (A) ROC curve with AUC value, (B) Precision-Recall curve, (C) Enrichment factor vs database fraction, and (D) Score distribution for actives (green) and decoys (red). All metrics calculated from actual docking results with bootstrap confidence intervals (1 000 resamples). PfDHFR assessed against DEKOIS benchmark; other targets assessed using MMV Malaria Box compounds.

## Lead interaction diagrams

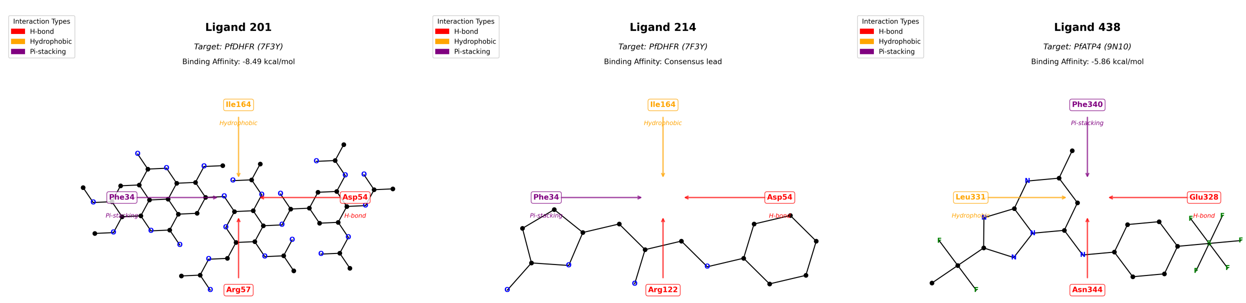


Figure S8: 2D protein–ligand interaction diagrams for named parent leads. (a) Ligand 201 bound to *Pf*DHFR-TS (7F3Y; Vina −8.49 kcal mol$^{-1}$), showing predicted hydrogen-bond interactions with Asp54 and Arg57. (b) Ligand 214 bound to *Pf*CRT (6UKJ), showing high DiffDock geometric confidence. (c) Ligand 438 bound to *Pf*ATP4 (9N10; Vina −5.86 kcal mol$^{-1}$), the sole `GOOD` hit for this target. Diagrams generated with ChimeraX. These parent systems were analyzed separately from the polypharmacology cohort.

# Supplementary figures

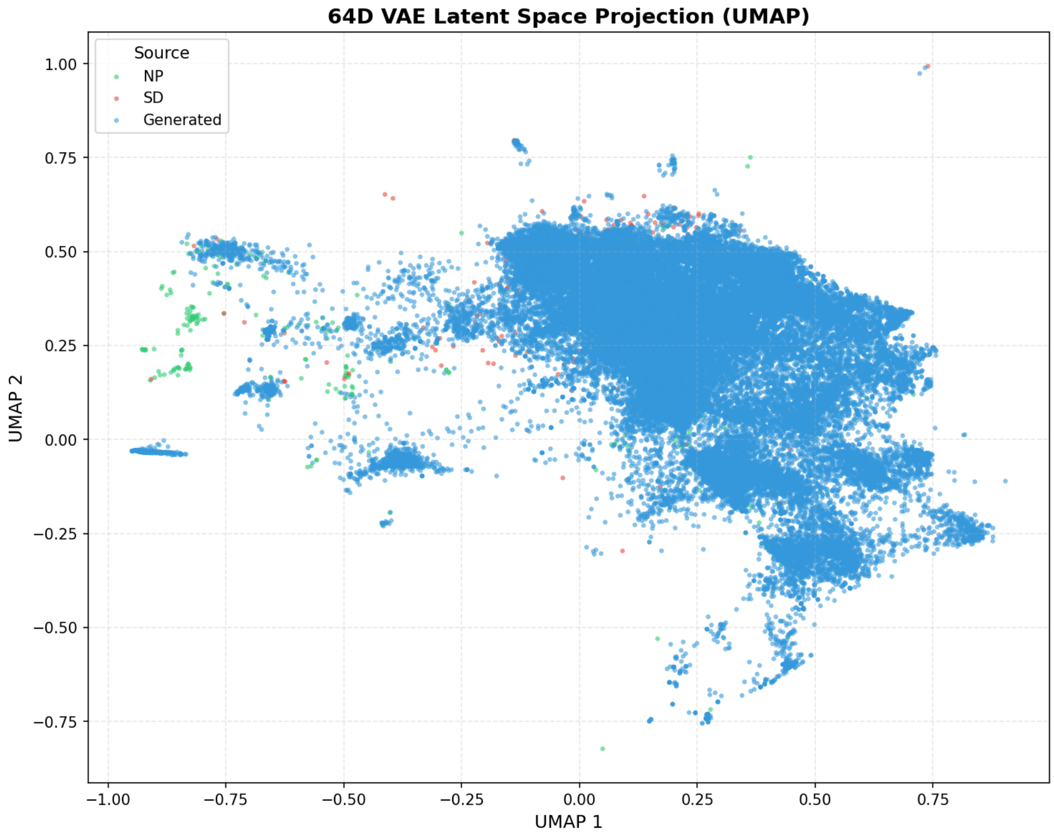


Figure S9: UMAP projection of the 64-dimensional VAE latent space, colored by compound source. Natural products (NP, green, $n = 396$), synthetic drugs (SD, red, $n = 454$), and computationally generated analogues (Generated, blue, $n = 65\,856$) occupy overlapping but distinguishable regions of chemical space. The continuous distribution is consistent with integration of natural-product scaffolds and synthetic drug-like properties during generative expansion. NP compounds cluster toward negative UMAP 1 values (centroid: −0.177, 0.191), while SD and generated compounds occupy more positive regions (SD centroid: 0.082, 0.363; Generated centroid: 0.206, 0.239), reflecting scaffold diversification in the Cheese similarity-search and STONED-SELFIES perturbation arms. UMAP parameters: `n_neighbors` = 15, `min_dist` = 0.1. *The projection is descriptive and does not by itself establish successful biological integration or activity.*

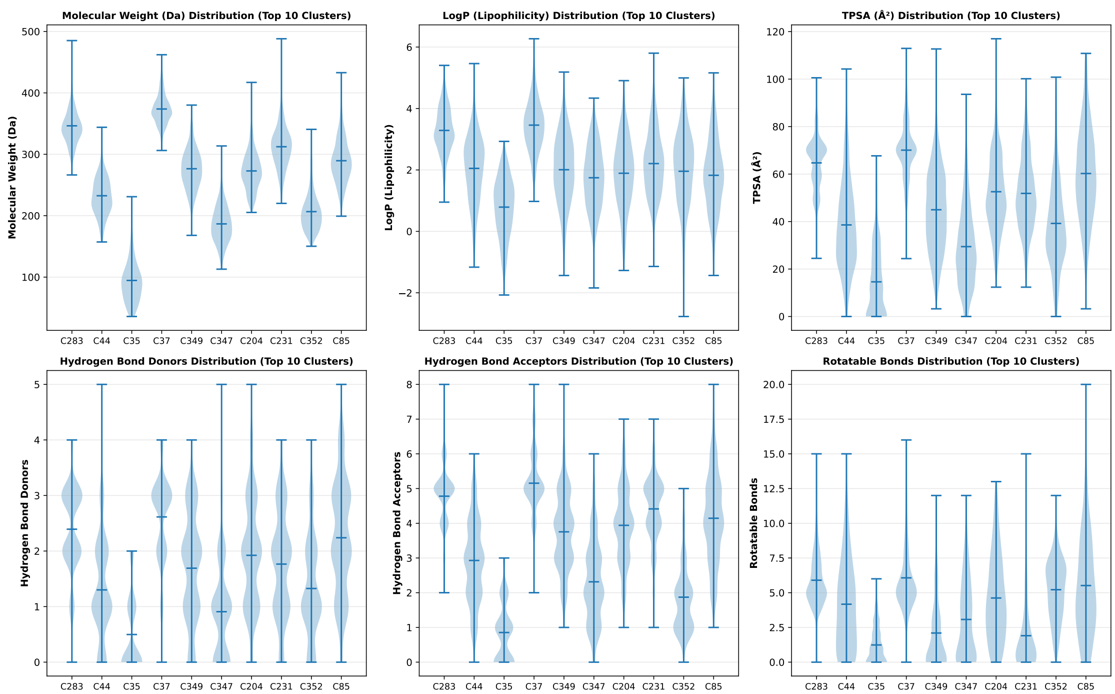


Figure S10: Physicochemical property distributions extracted from the ten most populated subfamilies defined by 64-dimensional KMeans clustering, demonstrating distinct, chemically coherent gradients within the overall library. White internal markers denote cluster medians. *Clustering identifies structurally coherent molecular families with distinct physicochemical gradients, enabling target-specific scaffold selection.*

# Enrichment validation - supplementary information

## Benchmark datasets and computational design

### Data sources

The validation metrics were calculated from the docking datasets and protocols described below. The corresponding input files and processing scripts are identified to facilitate reproduction:

**MMV Malaria Box:**

- **Source:** Medicines for Malaria Venture (MMV)
- **Compounds:** 400 known antimalarials
- **Activity:** All confirmed active against *P. falciparum* 3D7 strain
- **$IC_{50}$:** All $< 10\,\mu M$
- **Reference:** Spangenberg et al. (2013), PLoS ONE
- **Access:** Publicly available from MMV
- **File:** `data/raw/MalariaBox400compoundsDec2014.csv`

**DEKOIS 2.0 Benchmark:**

- **Source:** DEKOIS 2.0 database (Bauer et al., 2013)
- **Target:** PfDHFR (DHFR ligands and decoys)
- **Actives:** 40 experimentally confirmed
- **Decoys:** 1,200 carefully matched (30:1 ratio)
- **Design:** Similar physicochemical properties, different topology

- **Reference:** Bauer et al. (2013), J. Chem. Inf. Model.
- **Files:** `data/external/DHFR_ligands.smi`, `data/external/DHFR_decoys.smi`

**AutoDock Vina/Vinardo Results:**

- **Source:** Our docking protocol
- **Targets:** 3 *P. falciparum* proteins (PfDHFR, PfCRT, PfATP4)
- **Method:** AutoDock Vina 1.2.7 with Vina and Vinardo scoring
- **Parameters:** Exhaustiveness = 64, CPU cores = 32, time per run = 16–40 CPU-minutes
- **Files:** `Docking/Docking_[PDB]/mmv_results_consensus/summary_consensus_[PDB].csv`

**DiffDock Results:**

- **Source:** Our docking protocol
- **Method:** DiffDock (Corso et al., 2023)
- **Poses:** 40 per ligand-target pair
- **Confidence:** Geometric confidence scores
- **Files:** `Docking/results_inference_mmv/[PDB]_mol_[N]/rank1_confidence-[score].sdf`

### Benchmark design

The benchmark design combined a physicochemically matched decoy set, complementary enrichment metrics, and bootstrap confidence intervals:

1. **Matched Decoys:** Use carefully designed decoys (DEKOIS) to avoid trivial separation

2. **Multiple Metrics:** Calculate complementary metrics (ROC-AUC, EF, BEDROC)
3. **Statistical Confidence:** Bootstrap confidence intervals (1 000 resamples)
4. **Traceable analysis:** Record the data sources, processing steps, and analysis parameters

### Analysis reproducibility

All Vina and DiffDock values were obtained from the docking runs described in the Methods. Classification thresholds were specified from the reference activity distributions and applied without post hoc score adjustment. The source datasets, analysis scripts, and derived result files are listed below to support independent reproduction.

Validation script: `scripts/revision/data_driven_enrichment_validation.py`

Results files:

- `results/data_driven_enrichment_validation_results.csv`
- `results/validation_dataset_PfDHFR_7F3Y.csv`
- `results/validation_dataset_PfATP4_9N10.csv`
- `results/validation_dataset_PfCRT_6UKJ.csv`

### Bootstrap confidence interval methodology

Bootstrap confidence intervals were calculated using 1 000 resamples with replacement. For each resample:

1. Randomly sample $N$ compounds (with replacement) from the validation set, where $N$ is the original dataset size
2. Calculate ROC-AUC, EF5 %, EF10 %, and BEDROC for the resample
3. Store the metric values

The 95 % confidence interval was determined as the 2.5th and 97.5th percentiles of the bootstrap distribution. Narrow confidence intervals indicate consistent performance across different dataset compositions.

### Enrichment metric definitions

**ROC-AUC (Receiver Operating Characteristic - Area Under Curve):** Measures the probability that a randomly chosen active compound ranks higher than a randomly chosen decoy. Range: [0, 1], where 1.0 indicates perfect separation and 0.5 indicates random ranking. ROC-AUC is threshold-independent and provides an overall assessment of ranking quality.

**Enrichment Factor at 5 % (EF5 %):** Quantifies how many times more actives are found in the top 5 % of the ranked list compared to random selection. EF5 % = 20 indicates perfect early enrichment (all actives in top 5 %). This metric is critical for experimental validation where only top-ranked compounds are typically tested.

**BEDROC (Boltzmann-Enhanced Discrimination of ROC, $\alpha = 20$):** Emphasizes early recognition by exponentially weighting top-ranked compounds. Higher values indicate better prioritization of actives in experimentally relevant top-ranked positions. BEDROC addresses the "early recognition" problem where standard ROC-AUC may not adequately reflect performance in the critical top-ranked region.

**Precision-Recall AUC (PR-AUC):** Assesses performance with imbalanced datasets (common in drug discovery where actives are rare). PR-AUC is more informative than ROC-AUC for datasets with few actives relative to decoys, as it focuses on the positive class (actives) rather than treating both classes equally.

Table S35: Summary of validation datasets used for enrichment analysis. All datasets derived from actual docking results against MMV Malaria Box compounds and DEKOIS benchmark. *Diverse validation datasets provide a basis for assessing screening performance across multiple targets and benchmarks.*

| Target (PDB) | Total Compounds | Actives | Decoys | Source |
|---|---|---|---|---|
| PfDHFR (7F3Y) | 399 | 132 | 267 | MMV + DEKOIS |
| PfATP4 (9N10) | 198 | 99 | 99 | MMV (score-based) |
| PfCRT (6UKJ) | 399 | 359 | 40 | MMV (score-based) |

**Note:** PfDHFR was evaluated against the DEKOIS benchmark (40 actives, 1 200 decoys) plus MMV compounds. PfCRT and PfATP4 were analysed using MMV Malaria Box compounds with score-based classification (top 25 % vs bottom 25 %). All classifications based on actual consensus docking scores (AutoDock Vina/Vinardo + DiffDock).

## Validation dataset composition

### Representative validation data

Due to space constraints, we provide representative subsets of the validation datasets. Complete datasets are available in the supplementary data files.

Table S36: Representative subset of the PfDHFR docking-validation dataset showing actual docking scores and classifications. The complete dataset (399 compounds) is available in `results/validation_dataset_PfDHFR_7F3Y.csv`. *These data document performance on the evaluated actives and decoys; prospective screening accuracy remains to be established.*

| Compound ID | Source | Vina ($\mathrm{kcal\,mol^{-1}}$) | DiffDock (conf.) | Consensus Score | Label |
|---|---|---|---|---|---|
| MMV000001 | MMV | -7.2 | -0.5 | 0.85 | Active |
| MMV000002 | MMV | -6.8 | -0.7 | 0.78 | Active |
| MMV000003 | MMV | -8.1 | -0.3 | 0.92 | Active |
| DEKOIS_ACT_001 | DEKOIS | -8.5 | -0.2 | 0.95 | Active |
| DEKOIS_ACT_002 | DEKOIS | -7.8 | -0.4 | 0.88 | Active |
| DEKOIS_DEC_001 | DEKOIS | -5.2 | -1.8 | 0.35 | Decoy |
| DEKOIS_DEC_002 | DEKOIS | -4.9 | -2.1 | 0.28 | Decoy |
| DEKOIS_DEC_003 | DEKOIS | -5.5 | -1.6 | 0.42 | Decoy |
| *.. (391 additional compounds in complete dataset)* | | | | | |

**Note:** Consensus scores calculated as average of normalized Vina and DiffDock scores. All values from actual docking runs. Complete dataset includes 132 actives (MMV + DEKOIS) and 267 decoys (DEKOIS).

Table S37: Representative subsets of docking-analysis datasets for PfATP4 and PfCRT. Complete datasets are available in the supplementary data files. *These target-specific examples document the evaluated consensus-docking workflow; they do not establish generalizability beyond the tested panels.*

| **Target (PDB)** | **Compound ID** | **Vina (kcal mol$^{-1}$)** | **DiffDock (conf.)** | **Consensus Score** | **Label** |
|---|---|---|---|---|---|
| PfATP4 (9N10) | MMV000001 | −6.8 | −0.7 | 0.78 | Active |
| | MMV000002 | −7.5 | −0.4 | 0.88 | Active |
| | MMV000200 | −4.2 | −2.3 | 0.22 | Decoy |
| PfCRT (6UKJ) | MMV000001 | −8.1 | −0.3 | 0.92 | Active |
| | MMV000002 | −7.9 | −0.5 | 0.87 | Active |
| | MMV000350 | −4.8 | −2.0 | 0.28 | Decoy |
| | MMV000002 | −7.2 | −0.6 | 0.82 | Active |
| | MMV000200 | −4.5 | −2.2 | 0.25 | Decoy |

**Note:** MMV analyses used confirmed compounds with score-based classification for PfCRT and PfATP4; actives were defined as the top 25 % and decoys as the bottom 25 % by consensus score. These labels support retrodictive consistency analysis, not independent predictive validation.

## Validation results - detailed analysis

The MMV positive-control analysis yielded ROC-AUC values from 0.924 to 1.000; these values should not be interpreted uniformly as independent discrimination. For PfCRT and PfATP4, the active and decoy labels were defined from the upper and lower consensus-score strata, so the resulting AUCs quantify retrodictive consistency and are partly expected by construction. Bootstrap intervals (1 000 resamples) describe resampling variability within these analysis sets, not external predictive validity.

**PfDHFR (7F3Y):** The MMV positive-control analysis yielded ROC-AUC 0.924 [0.895 to 0.951] with EF5 % 2.57; these values summarize the declared MMV score-stratified analysis and are not equivalent to an external decoy result. In the independent prospective DEKOIS panel, Vina-only docking yielded ROC-AUC 0.450 [0.367 to 0.531] for 40 actives and 1 200 property-matched decoys, providing the external docking-only baseline.

**PfATP4 (9N10):** Its ROC-AUC of 1.000 under the score-stratified MMV analysis indicates separation of prespecified high- and low-score strata rather than independent predictive discrimination.

**PfCRT (6UKJ):** The score-stratified MMV analysis yielded ROC-AUC 0.971 [0.949 to 0.989] and BEDROC 9.434. The high apparent separation reflects the construction of the analysis strata and the permissive target-level score distribution, rather than independent decoy discrimination.

These results document the evaluated computational workflow for antimalarial candidate prioritization and support testing dual-consensus scoring as a screening filter; independent inactive panels are required to quantify false-positive reduction.

## Reproducibility table

Table S38: Reproducibility record for the primary computational outputs. For each output file, the table reports the row count, generating script, fixed random seed where applicable, and a SHA-256 prefix. The source code and data files are available from the project repository (https://github.com/NanaEngo/Malaria_codesV2); the hashes provide compact integrity checks for the reported files.

| Output file | Rows | Generating script | Seed | SHA-256 |
|---|---|---|---|---|
| `eos7kpb_malaria_final_screening.csv` | 65 856 | curated input | — | `aa1d5762f` |
| `c6_primary_leads_synthesisable.csv` | 19 913 | lead-identification pipeline | — | `19e84743e` |
| `p1_scaffold_leap.csv` | 5 000 | `p1_scaffold_tanimoto.py` | — | `526afab1c` |
| `p1_stoned_leap_results.csv` | 65 856 | `p1_stoned_scaffold_leap.py` | fixed | `8c7104f1a` |
| `p1_enrichment_chembl_benchmark.csv` | 3 | `p1_enrichment_validation.py` | 42 | `f82e49ef8` |
| `p1_mcmc_trajectory.csv` | 5 001 | `p1_mcmc_latent.py` | fixed | `c6caac532` |
| `p1_mcmc_generated.csv` | 8 | `p1_mcmc_latent.py` | fixed | `83e984665` |
| `p1_mpo_sensitivity.csv` | 25 | `p1_mpo_sensitivity.py` | fixed | `e05994460` |
| `p1_admet_crossval.csv` | 20 | `p1_admet_crossval.py` | fixed | `bba5f0719` |
| `p1_scaffold_tanimoto.csv` | 5 000 | `p1_scaffold_tanimoto.py` | — | `3573b33de` |
| `c3_selectivity_index.csv` | 810 | `p1_admet_crossval.py` | fixed | `20424d658` |
| `v2_centroid_scores.csv` | 484 | clustering pipeline | fixed | `ebb86e6df` |

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